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31.

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Rules for the Direction of the Mind
Discourse on the Method
Meditations on First Philosophy
Objections against the Meditations and Replies
The Geometry

BY RENÉ DESCARTES

Ethics

BY BENEDICT DE SPINOZA



WILLIAM BENTON, *Publisher*

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DESCARTES

BIOGRAPHICAL NOTE

René Descartes, 1596–1650

Descartes by birth belonged to the lesser nobility; both of his parents came from high legal families. He was born at La Haye, in Touraine, on March 31, 1596. Although a younger son, he derived an income, sufficient to make him independent throughout his life, from the property left him by his mother.

While still a boy Descartes was sent to the Jesuit School at La Flêche, founded by Henry IV and "one of the most celebrated schools in Europe." He appears to have been at La Flêche from 1606 to 1615, following the Jesuit program of studies which aimed at reconciling the classical learning of the Renaissance with the scholastic philosophy of the Middle Ages. Suffering from poor health, he was entrusted to the special care of Father Dinet, afterwards the confessor to Louis XIII and Louis XIV. He was excused from morning duties and allowed to stay in bed, a habit he retained to the end of his life. After completing the full curriculum of languages and humane letters, logic, ethics, mathematics, physics, and metaphysics, Descartes later declared, "I found myself embarrassed with so many doubts and errors, that it seemed to me that the effort to instruct myself had no effect other than the increasing discovery of my own ignorance." Mathematics alone appeared to be an exception "because of the certainty of its demonstrations and the evidence of its reasoning." He completed his education at the University of Poitiers, where he took his degree in law, November 10, 1616.

Descartes spent the remainder of his youth in travelling, "resolved no longer to seek any other science than the knowledge of myself, or of the great book of the world." Like many young Frenchmen of the time, he enlisted as a gentleman volunteer in the army of Prince Maurice of Nassau in Holland. He was still interested in mathematics, and at Breda became a friend of Isaac Beeckman, mathematician and rector of the college at Dort. Beeckman, after their meeting, noted in his diary, "Mathematical physicists are scarce, and I myself had never had any conversation on that topic with anybody but him." Their discussions, according to Descartes, turned his mind to purely theoretical problems, and when he left Holland early in 1619 to seek more active military service in Germany, he had already completed an *Essay on Algebra* and a *Compendium on Music*, dedicated to his friend.

Descartes dated his life as a philosopher from 1619. Early in that year, after his study of algebra and geometry had yielded what he considered an "entirely new science," he wrote to his Dutch friend: "My project is unbelievably ambitious, but I cannot help feeling that I am sighting I know not what light in the chaos of present-day geometry, and I trust that it will help me in dispelling that most opaque darkness." In the fall, after the army had gone into winter quarters, he retired to a village near Ulm on the Danube to devote himself to study and speculation. "On November 10, 1619," he wrote, "when I was filled with enthusiasm, I discovered the foundations of the wonderful science." The discovery was followed by a series of three dreams which left Descartes the impression that "the Spirit of Truth had opened to him the treasures of all the sciences."

The experience of November 10 did not immediately alter his way of life. Some time previously he had remarked, "As comedians put on a mask to hide their timidity, so I go forward masked preparing to mount the stage of the world, which up to now I have known only as a spectator"; and for the next nine years he continued to live as a soldier and a "gentilhomme" while preparing to apply his newly discovered method to all knowledge. In 1622 he was back in France, frequenting the society of the leading scientists and philosophers. Through his friends and correspondence he was already known and esteemed for his scientific abilities, although he had not as yet published anything. He appears to have been reluctant to make his work public until his researches in physics promised to yield practical results, and he felt he could no longer "keep them concealed without greatly sinning against the law which obliges us to procure, as far as in us lies, the general good of all mankind." At the same time he had occasion to discuss his research with Cardinal Bérulle, who was so impressed that he declared

Descartes was morally obliged to make his thought known to the world. Feeling that he could not find in Paris the leisure and quiet he needed for writing, Descartes retired to Holland.

From 1629 until 1649 Descartes lived in Holland, leaving only for five short visits, three to France, one to England, and another to Denmark. He disliked dwelling for long in the same place and during that time changed his residence twenty-four times, concerned only, it would appear, to be in the neighborhood of a university and a Catholic church. Most of his more important works were written and published in Holland. He wrote the *Rules for the Direction of the Mind* during the first year, and by 1633 had all but completed his *Treatise on the World*, when the condemnation of Galileo caused him to abandon all thought of publishing it. In 1637 he brought out the *Discourse on Method* with the three "Essays" accompanying it, the *Dioptric*, *Meteors*, and *Geometry*. Through Mersenne, who acted as his personal secretary in Paris, he circulated a manuscript of his *Meditations* and obtained objections to its arguments; the work was published with his answers to the objections in 1641.

Descartes' philosophy became a source of controversy in Holland even before the appearance of his works, as a result of the teaching of his friends in the universities. Cartesianism was attacked as subversive of religion, and at one time Descartes was summoned before the magistrates of Utrecht, although the matter went no further because of the intervention of influential friends.

Among his friends and admirers was Princess Elizabeth, daughter of Emperor Frederick V, then in exile in Holland. Although she was only nineteen when the *Discourse* appeared, she was interested in philosophical discussion, and Descartes, in dedicating the *Principles of Philosophy* (1644) to her, declared that hers was "the only mind, as far as my experience goes, to which both metaphysics and mathematics are easy." Queen Christina of Sweden also became interested in the "new philosophy," and, through the French Ambassador, Descartes carried on a correspondence with her on ethical subjects, part of which was reworked and published as the *Treatise on the Passions of the Soul* (1650). Late in 1649 she persuaded Descartes to go to the Swedish court. He was charged with the task of drawing up a statute for a proposed academy of science and teaching philosophy to the Queen. The lessons in philosophy were scheduled to be given three times a week at five in the morning. Descartes contracted an inflammation of the lungs and died after a very brief illness, February 11, 1650.

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RULES
FOR THE DIRECTION OF THE MIND

RULE I

The end of study should be to direct the mind towards the enunciation of sound and correct judgments on all matters that come before it.

WHENEVER men notice some similarity between two things, they are wont to ascribe to each, even in those respects in which the two differ, what they have found to be true of the other. Thus they erroneously compare the sciences, which entirely consist in the cognitive exercise of the mind, with the arts, which depend upon an exercise and disposition of the body. They see that not all the arts can be acquired by the same man, but that he who restricts himself to one, most readily becomes the best executant, since it is not so easy for the same hand to adapt itself both to agricultural operations and to harp-playing, or to the performance of several such tasks as to one alone. Hence they have held the same to be true of the sciences also, and distinguishing them from one another according to their subject matter, they have imagined that they ought to be studied separately, each in isolation from all the rest. But this is certainly wrong. For since the sciences taken all together are identical with human wisdom, which always remains one and the same, however applied to different subjects, and suffers no more differentiation proceeding from them than the light of the sun experiences from the variety of the things which it illumines, there is no need for minds to be confined at all within limits; for neither does the knowing of one truth have an effect like that of the acquisition of one art and prevent us from finding out another, it rather aids us to do so. Certainly it appears to me strange that so many people should investigate human customs with such care, the virtues of plants, the motions of the stars, the transmutations of metals, and the objects of similar sciences, while at the same time practically none bethink themselves about good understanding, or universal Wisdom, though nevertheless all other studies are to be esteemed not so much for their own value as because they contribute something to this. Consequently we are justified in bringing forward this as the first rule of all, since there is nothing more prone to turn us aside from the correct way of seeking out truth than this directing of our inquiries, not towards their general end, but towards certain special investigations. I do not here refer to perverse and censurable pursuits like empty glory or base gain; obviously counterfeit reasonings and quibbles suited to vulgar understanding open up a much more direct route to such a goal than does a sound apprehension of the truth. But I have in view even honourable and laudable pursuits, because these mislead us in a more subtle fashion. For example take our investigations of those sciences conducive to the conveniences of life or which yield that pleasure which is found in the contemplation of truth, practically the only joy in life that is complete and untroubled with any pain. There we may indeed expect to receive the legitimate fruits of scientific inquiry; but if, in the course of our study, we think of them, they frequently cause us to omit many facts which are necessary to the understanding of other matters, because they seem to be either of slight value or of little interest. Hence we must believe that all the sciences are so inter-connected, that it is much easier to study them all together than to isolate one from all the others. If, therefore, anyone wishes to search out the truth of things in serious earnest, he ought not to select one special science; for all the sciences are conjoined with each other and interdependent: he ought rather to think how to increase the natural light of reason, not for the purpose of resolving this or that difficulty of scholastic type, but in order that his understanding may light his will to its proper choice in all the contin-

gencies of life. In a short time he will see with amazement that he has made much more progress than those who are eager about particular ends, and that he has not only obtained all that they desire, but even higher results than fall within his expectation.

RULE II

Only those objects should engage our attention, to the sure and indubitable knowledge of which our mental powers seem to be adequate.

SCIENCE in its entirety is true and evident cognition. He is no more learned who has doubts on many matters than the man who has never thought of them; nay he appears to be less learned if he has formed wrong opinions on any particulars. Hence it were better not to study at all than to occupy one's self with objects of such difficulty, that, owing to our inability to distinguish true from false, we are forced to regard the doubtful as certain; for in those matters any hope of augmenting our knowledge is exceeded by the risk of diminishing it. Thus in accordance with the above maxim we reject all such merely probable knowledge and make it a rule to trust only what is completely known and incapable of being doubted. No doubt men of education may persuade themselves that there is but little of such certain knowledge, because, forsooth, a common failing of human nature has made them deem it too easy and open to everyone, and so led them to neglect to think upon such truths; but I nevertheless announce that there are more of these than they think—truths which suffice to give a rigorous demonstration of innumerable propositions, the discussion of which they have hitherto been unable to free from the element of probability. Further, because they have believed that it was unbecoming for a man of education to confess ignorance on any point, they have so accustomed themselves to trick out their fabricated explanations, that they have ended by gradually imposing on themselves and thus have issued them to the public as genuine.

But if we adhere closely to this rule we shall find left but few objects of legitimate study. For there is scarce any question occurring in the sciences about which talented men have not disagreed. But whenever two men come to opposite decisions about the same matter one of them at least must certainly be in the wrong, and apparently there is not even one of them who knows; for if the reasoning of the second was sound and clear he would be able so to lay it before the other as finally to succeed in convincing *his* understanding also. Hence apparently we cannot attain to a perfect knowledge in any such case of probable opinion, for it would be rashness to hope for more than others have attained to. Consequently if we reckon correctly, of the sciences already discovered, Arithmetic and Geometry alone are left, to which the observance of this rule reduces us.

Yet we do not therefore condemn that method of philosophizing which others have already discovered and those weapons of the schoolmen, probable syllogisms, which are so well suited for polemics. They indeed give practice to the wits of youths and, producing emulation among them, act as a stimulus; and it is much better for their minds to be moulded by opinions of this sort, uncertain though they appear, as being objects of controversy among the learned, than to be left entirely to their own devices. For thus through lack of guidance they might stray into some abyss; but as long as they follow in their masters' footsteps, though they may diverge at times from the truth, they will yet certainly find a path which is at least in this respect safer, that it has been approved of by more prudent people. We ourselves rejoice that we in earlier years experienced this scholastic training; but now, being released from that oath of allegiance which bound us to our old masters, and since, as becomes our riper years, we are no longer subject to the ferule, if we wish in earnest to establish for ourselves those rules which shall aid us in scaling the heights of human knowledge, we must admit assuredly among the primary members of our catalogue that maxim which forbids us to abuse our leisure as many do, who neglect all easy quests and take up their time only with difficult matters; for they, though certainly making all sorts of subtle conjectures and elaborating most plausible arguments with great ingenuity, frequently find too late that after all their labours they have only increased the multitude of their doubts, without acquiring any knowledge whatsoever.

But now let us proceed to explain more carefully our reasons for saying, as we did a little while ago, that of all the sciences known as yet, Arithmetic and Geometry alone are free from any taint of falsity or uncertainty. We must note then that there are two ways by which we arrive at the knowledge of facts, viz. by experience and by deduction. We must further observe that while our inferences from experience are frequently fallacious, deduction, or the pure illation of one thing from another,

though it may be passed over, if it is not seen through, cannot be erroneous when performed by an understanding that is in the least degree rational. And it seems to me that the operation is profited but little by those constraining bonds by means of which the Dialecticians claim to control human reason, though I do not deny that that discipline may be serviceable for other purposes. My reason for saying so is that none of the mistakes which men can make (men, I say, not beasts) are due to faulty inference; they are caused merely by the fact that we found upon a basis of poorly comprehended experiences, or that propositions are posited which are hasty and groundless.

This furnishes us with an evident explanation of the great superiority in certitude of Arithmetic and Geometry to other sciences. The former alone deal with an object so pure and uncomplicated, that they need make no assumptions at all which experience renders uncertain, but wholly consist in the rational deduction of consequences. They are on that account much the easiest and clearest of all, and possess an object such as we require, for in them it is scarce humanly possible for anyone to err except by inadvertence. And yet we should not be surprised to find that plenty of people of their own accord prefer to apply their intelligence to other studies, or to Philosophy. The reason for this is that every person permits himself the liberty of making guesses in the matter of an obscure subject with more confidence than in one which is clear, and that it is much easier to have some vague notion about any subject, no matter what, than to arrive at the real truth about a single question however simple that may be.

But one conclusion now emerges out of these considerations, viz. not, indeed, that Arithmetic and Geometry are the sole sciences to be studied, but only that in our search for the direct road towards truth we should busy ourselves with no object about which we cannot attain a certitude equal to that of the demonstrations of Arithmetic and Geometry.

RULE III

In the subjects we propose to investigate, our inquiries should be directed, not to what others have thought, nor to what we ourselves conjecture, but to what we can clearly and perspicuously behold and with certainty deduce; for knowledge is not won in any other way.

To STUDY the writings of the ancients is right, because it is a great boon for us to be able to make use of the labours of so many men; and we should do so, both in order to discover what they have correctly made out in previous ages, and also that we may inform ourselves as to what in the various sciences is still left for investigation. But yet there is a great danger lest in a too absorbed study of these works we should become infected with their errors, guard against them as we may. For it is the way of writers, whenever they have allowed themselves rashly and credulously to take up a position in any controverted matter, to try with the subtlest of arguments to compel us to go along with them. But when, on the contrary, they have happily come upon something certain and evident, in displaying it they never fail to surround it with ambiguities, fearing, it would seem, lest the simplicity of their explanation should make us respect their discovery less, or because they grudge us an open vision of the truth.

Further, supposing now that all were wholly open and candid, and never thrust upon us doubtful opinions as true, but expounded every matter in good faith, yet since scarce anything has been asserted by any one man the contrary of which has not been alleged by another, we should be eternally uncertain which of the two to believe. It would be no use to total up the testimonies in favour of each, meaning to follow that opinion which was supported by the greater number of authors: for if it is a question of difficulty that is in dispute, it is more likely that the truth would have been discovered by few than by many. But even though all these men agreed among themselves, what they teach us would not suffice for us. For we shall not, e.g. all turn out to be mathematicians though we know by heart all the proofs that others have elaborated, unless we have an intellectual talent that fits us to resolve difficulties of any kind. Neither, though we have mastered all the arguments of Plato and Aristotle, if yet we have not the capacity for passing a solid judgment on these matters, shall we become Philosophers; we should have acquired the knowledge not of a science, but of history.

I lay down the rule also, that we must wholly refrain from ever mixing up conjectures with our pronouncements on the truth of things. This warning is of no little importance. There is no stronger reason for our finding nothing in the current Philosophy which is so evident and certain as not to be capable of being controverted, than the fact that the learned, not con-

tent with the recognition of what is clear and certain, in the first instance hazard the assertion of obscure and ill-comprehended theories, at which they have arrived merely by probable conjecture. Then afterwards they gradually attach complete credence to them, and mingling them promiscuously with what is true and evident, they finish by being unable to deduce any conclusion which does not appear to depend upon some proposition of the doubtful sort, and hence is not uncertain.

But lest we in turn should slip into the same error, we shall here take note of all those mental operations by which we are able, wholly without fear of illusion, to arrive at the knowledge of things. Now I admit only two, viz. intuition and induction.

By *intuition* I understand, not the fluctuating testimony of the senses, nor the misleading judgment that proceeds from the blundering constructions of imagination, but the conception which an unclouded and attentive mind gives us so readily and distinctly that we are wholly freed from doubt about that which we understand. Or, what comes to the same thing, *intuition* is the undoubting conception of an unclouded and attentive mind, and springs from the light of reason alone; it is more certain than deduction itself, in that it is simpler, though deduction, as we have noted above, cannot by us be erroneously conducted. Thus each individual can mentally have intuition of the fact that he exists, and that he thinks; that the triangle is bounded by three lines only, the sphere by a single superficies, and so on. Facts of such a kind are far more numerous than many people think, disdaining as they do to direct their attention upon such simple matters.

But in case anyone may be put out by this new use of the term *intuition* and of other terms which in the following pages I am similarly compelled to dissever from their current meaning, I here make the general announcement that I pay no attention to the way in which particular terms have of late been employed in the schools, because it would have been difficult to employ the same terminology while my theory was wholly different. All that I take note of is the meaning of the Latin of each word, when, in cases where an appropriate term is lacking, I wish to transfer to the vocabulary that expresses my own meaning those that I deem most suitable.

This evidence and certitude, however, which belongs to intuition, is required not only in the enunciation of propositions, but also in discursive reasoning of whatever sort. For example consider this consequence: 2 and 2 amount to the same as 3 and 1. Now we need to see intuitively not only that 2 and 2 make 4, and that likewise 3 and 1 make 4, but further that the third of the above statements is a necessary conclusion from these two.

Hence now we are in a position to raise the question as to why we have, besides intuition, given this supplementary method of knowing, viz. knowing by *deduction*, by which we understand all necessary inference from other facts that are known with certainty. This, however, we could not avoid, because many things are known with certainty, though not by themselves evident, but only deduced from true and known principles by the continuous and uninterrupted action of a mind that has a clear vision of each step in the process. It is in a similar way that we know that the last link in a long chain is connected with the first, even though we do not take in by means of one and the same act of vision all the intermediate links on which that connection depends, but only remember that we have taken them successively under review and that each single one is united to its neighbour, from the first even to the last. Hence we distinguish this mental intuition from deduction by the fact that into the conception of the latter there enters a certain movement or succession, into that of the former there does not. Further deduction does not require an immediately presented evidence such as intuition possesses; its certitude is rather conferred upon it in some way by memory. The upshot of the matter is that it is possible to say that those propositions indeed which are immediately deduced from first principles are known now by intuition, now by deduction, i.e. in a way that differs according to our point of view. But the first principles themselves are given by intuition alone, while, on the contrary, the remote conclusions are furnished only by deduction.

These two methods are the most certain routes to knowledge, and the mind should admit no others. All the rest should be rejected as suspect of error and dangerous. But this does not prevent us from believing matters that have been divinely revealed as being more certain than our surest knowledge, since belief in these things, as all faith in obscure matters, is an action not of our intelligence, but of our will. They should be heeded also since, if they have any basis in our understanding, they can

and ought to be, more than all things else, discovered by one or other of the ways above-mentioned, as we hope perhaps to show at greater length on some future opportunity.

RULE IV

There is need of a method for finding out the truth.

So blind is the curiosity by which mortals are possessed, that they often conduct their minds along unexplored routes, having no reason to hope for success, but merely being willing to risk the experiment of finding whether the truth they seek lies there. As well might a man burning with an unintelligent desire to find treasure, continuously roam the streets, seeking to find something that a passer by might have chanced to drop. This is the way in which most Chemists, many Geometricians, and Philosophers not a few prosecute their studies. I do not deny that sometimes in these wanderings they are lucky enough to find something true. But I do not allow that this argues greater industry on their part, but only better luck. But however that may be, it were far better never to think of investigating truth at all, than to do so without a method. For it is very certain that unregulated inquiries and confused reflections of this kind only confound the natural light and blind our mental powers. Those who so become accustomed to walk in darkness weaken their eye-sight so much that afterwards they cannot bear the light of day. This is confirmed by experience; for how often do we not see that those who have never taken to letters, give a sounder and clearer decision about obvious matters than those who have spent all their time in the schools? Moreover by a method I mean certain and simple rules, such that, if a man observe them accurately, he shall never assume what is false as true, and will never spend his mental efforts to no purpose, but will always gradually increase his knowledge and so arrive at a true understanding of all that does not surpass his powers.

These two points must be carefully noted, viz. never to assume what is false as true, and to arrive at a knowledge which takes in all things. For, if we are without the knowledge of any of the things which we are capable of understanding, that is only because we have never perceived any way to bring us to this knowledge, or because we have fallen into the contrary error. But if our method rightly explains how our mental vision should be used, so as not to fall into the contrary error, and how deduction should be discovered in order that we may arrive at the knowledge of all things, I do not see what else is needed to make it complete; for I have already said that no science is acquired except by mental intuition or deduction. There is besides no question of extending it further in order to show how these said operations ought to be effected, because they are the most simple and primary of all. Consequently, unless our understanding were already able to employ them, it could comprehend none of the precepts of that very method, not even the simplest. But as for the other mental operations, which Dialectic does its best to direct by making use of these prior ones, they are quite useless here; rather they are to be accounted impediments, because nothing can be added to the pure light of reason which does not in some way obscure it.

Since then the usefulness of this method is so great that without it study seems to be harmful rather than profitable, I am quite ready to believe that the greater minds of former ages had some knowledge of it, nature even conducting them to it. For the human mind has in it something that we may call divine, wherein are scattered the first germs of useful modes of thought. Consequently it often happens that however much neglected and choked by interfering studies they bear fruit of their own accord. Arithmetic and Geometry, the simplest sciences, give us an instance of this; for we have sufficient evidence that the ancient Geometricians made use of a certain analysis which they extended to the resolution of all problems, though they grudged the secret to posterity. At the present day also there flourishes a certain kind of Arithmetic, called Algebra, which designs to effect, when dealing with numbers, what the ancients achieved in the matter of figures. These two methods are nothing else than the spontaneous fruit sprung from the inborn principles of the discipline here in question; and I do not wonder that these sciences with their very simple subject matter should have yielded results so much more satisfactory than others in which greater obstructions choke all growth. But even in the latter case, if only we take care to cultivate them assiduously, fruits will certainly be able to come to full maturity. — with great care & attention.

This is the chief result which I have had in view in writing this treatise. For I should not think much of these rules, if they had no utility save for the solution of the empty problems with which Logicians and Geometers have

been wont to beguile their leisure; my only achievement thus would have seemed to be an ability to argue about trifles more subtly than others. Further, though much mention is here made of numbers and figures, because no other sciences furnish us with illustrations of such self-evidence and certainty, the reader who follows my drift with sufficient attention will easily see that nothing is less in my mind than ordinary Mathematics, and that I am expounding quite another science, of which these illustrations are rather the outer husk than the constituents. Such a science should contain the primary rudiments of human reason, and its province ought to extend to the eliciting of true results in every subject. To speak freely, I am convinced that it is a more powerful instrument of knowledge than any other that has been bequeathed to us by human agency, as being the source of all others. But as for the outer covering I mentioned, I mean not to employ it to cover up and conceal my method for the purpose of warding off the vulgar; rather I hope so to clothe and embellish it that I may make it more suitable for presentation to the human mind.

When first I applied my mind to Mathematics I read straight away most of what is usually given by the mathematical writers, and I paid special attention to Arithmetic and Geometry, because they were said to be the simplest and so to speak the way to all the rest. But in neither case did I then meet with authors who fully satisfied me. I did indeed learn in their works many propositions about numbers which I found on calculation to be true. As to figures, they in a sense exhibited to my eyes a great number of truths and drew conclusions from certain consequences. But they did not seem to make it sufficiently plain to the mind itself why those things are so, and how they discovered them. Consequently I was not surprised that many people, even of talent and scholarship, should, after glancing at these sciences, have either given them up as being empty and childish or, taking them to be very difficult and intricate, been deterred at the very outset from learning them. For really there is nothing more futile than to busy one's self with bare numbers and imaginary figures in such a way as to appear to rest content with such trifles, and so to resort to those superficial demonstrations, which are discovered more frequently by chance than by skill, and are a matter more of the eyes and the imagination than of the understanding, that in a sense one ceases to make use of one's reason. I might add that there is no more intricate task than that of solving by this method of proof new difficulties that arise, involved as they are with numerical confusions. But when I afterwards bethought myself how it could be that the earliest pioneers of Philosophy in bygone ages refused to admit to the study of wisdom any one who was not versed in Mathematics, evidently believing that this was the easiest and most indispensable mental exercise and preparation for laying hold of other more important sciences, I was confirmed in my suspicion that they had knowledge of a species of Mathematics very different from that which passes current in our time. I do not indeed imagine that they had a perfect knowledge of it, for they plainly show how little advanced they were by the insensate rejoicings they display and the pompous thanksgivings they offer for the most trifling discoveries. I am not shaken in my opinion by the fact that historians make a great deal of certain machines of theirs. Possibly these machines were quite simple, and yet the ignorant and wonder-loving multitude might easily have lauded them as miraculous. But I am convinced that certain primary germs of truth implanted by nature in human minds—though in our case the daily reading and hearing of innumerable diverse errors stifle them—had a very great vitality in that rude and unsophisticated age of the ancient world. Thus the same mental illumination which let them see that virtue was to be preferred to pleasure, and honour to utility, although they knew not why this was so, made them recognize true notions in Philosophy and Mathematics, although they were not yet able thoroughly to grasp these sciences. Indeed I seem to recognize certain traces of this true Mathematics in Pappus and Diophantus, who though not belonging to the earliest age, yet lived many centuries before our own times. But my opinion is that these writers then with a sort of low cunning, deplorable indeed, suppressed this knowledge. Possibly they acted just as many inventors are known to have done in the case of their discoveries, i.e. they feared that their method being so easy and simple would become cheapened on being divulged, and they preferred to exhibit in its place certain barren truths, deductively demonstrated with show enough of ingenuity, as the results of their art, in order to win from us our admiration for these achievements, rather than to disclose to us that method itself which would

have wholly annulled the admiration accorded. Finally, there have been certain men of talent who in the present age have tried to revive this same art. For it seems to be precisely that science known by the barbarous name Algebra if only we could extricate it from that vast array of numbers and inexplicable figures by which it is overwhelmed, so that it might display the clearness and simplicity which, we imagine, ought to exist in a genuine Mathematics.

It was these reflections that recalled me from the particular studies of Arithmetic and Geometry to a general investigation of Mathematics, and thereupon I sought to determine what precisely was universally meant by that term, and why not only the above mentioned sciences, but also Astronomy, Music, Optics, Mechanics and several others are styled parts of Mathematics. Here indeed it is not enough to look at the origin of the word; for since the name "Mathematics" means exactly the same thing as "scientific study," these other branches could, with as much right as Geometry itself, be called Mathematics. Yet we see that almost anyone who has had the slightest schooling, can easily distinguish what relates to Mathematics in any question from that which belongs to the other sciences. But as I considered the matter carefully it gradually came to light that all those matters only were referred to Mathematics in which order and measurement are investigated, and that it makes no difference whether it be in numbers, figures, stars, sounds or any other object that the question of measurement arises. I saw consequently that there must be some general science to explain that element as a whole which gives rise to problems about order and measurement, restricted as these are to no special subject matter. This, I perceived, was called "Universal Mathematics," not a far fetched designation, but one of long standing which has passed into current use, because in this science is contained everything on account of which the others are called parts of Mathematics. We can see how much it excels in utility and simplicity the sciences subordinate to it, by the fact that it can deal with all the objects of which they have cognizance and many more besides, and that any difficulties it contains are found in them as well, added to the fact that in them fresh difficulties arise due to their special subject matter which in it do not exist. But now how comes it that though everyone knows the name of this science and understands what is its province even without studying it attentively, so many people laboriously pursue the other dependent sciences, and no one cares to master this one? I should marvel indeed were I not aware that everyone thinks it to be so very easy, and had I not long since observed that the human mind passes over what it thinks it can easily accomplish, and hastens straight away to new and more imposing occupations.

I, however, conscious as I am of my inadequacy, have resolved that in my investigation into truth I shall follow obstinately such an order as will require me first to start with what is simplest and easiest, and never permit me to proceed farther until in the first sphere there seems to be nothing further to be done. This is why up to the present time to the best of my ability I have made a study of this universal Mathematics; consequently, I believe that when I go on to deal in their turn with more profound sciences, as I hope to do soon, my efforts will not be premature. But before I make this transition I shall try to bring together and arrange in an orderly manner, the facts which in my previous studies I have noted as being more worthy of attention. Thus I hope both that at a future date, when through advancing years my memory is enfeebled, I shall, if need be, conveniently be able to recall them by looking in this little book, and that having now disburdened my memory of them I may be free to concentrate my mind on my future studies.

RULE V

Method consists entirely in the order and disposition of the objects towards which our mental vision must be directed if we would find out any truth. We shall comply with it exactly if we reduce involved and obscure propositions step by step to those that are simpler, and then starting with the intuitive apprehension of all those that are absolutely simple, attempt to ascend to the knowledge of all others by precisely similar steps.

IN THIS alone lies the sum of all human endeavour, and he who would approach the investigation of truth must hold to this rule as closely as he who enters the labyrinth must follow the thread which guided Theseus. But many people either do not reflect on the precept at all, or ignore it altogether, or presume not to need it. Consequently, they often investigate the most difficult questions with so little regard to order, that, to my mind, they act like a man who should attempt to leap with one bound from the base to the summit of a

house, either making no account of the ladders provided for his ascent or not noticing them. It is thus that all Astrologers behave, who, though in ignorance of the nature of the heavens, and even without having made proper observations of the movements of the heavenly bodies, expect to be able to indicate their effects. This is also what many do who study Mechanics apart from Physics, and rashly set about devising new instruments for producing motion. Along with them go also those Philosophers who, neglecting experience, imagine that truth will spring from their brain like Pallas from the head of Zeus.

Now it is obvious that all such people violate the present rule. But since the order here required is often so obscure and intricate that not everyone can make it out, they can scarcely avoid error unless they diligently observe what is laid down in the following proposition.

RULE VI

In order to separate out what is quite simple from what is complex, and to arrange these matters methodically, we ought, in the case of every series in which we have deduced certain facts the one from the other, to notice which fact is simple, and to mark the interval, greater, less, or equal, which separates all the others from this.

Although this proposition seems to teach nothing very new, it contains, nevertheless, the chief secret of method, and none in the whole of this treatise is of greater utility. For it tells us that all facts can be arranged in certain series, not indeed in the sense of being referred to some ontological genus such as the categories employed by Philosophers in their classification, but in so far as certain truths can be known from others; and thus, whenever a difficulty occurs we are able at once to perceive whether it will be profitable to examine certain others first, and which, and in what order.

Further, in order to do that correctly, we must note first that for the purpose of our procedure, which does not regard things as isolated realities, but compares them with one another in order to discover the dependence in knowledge of one upon the other, all things can be said to be either absolute or relative.

I call that absolute which contains within itself the pure and simple essence of which we are in quest. Thus the term will be applicable to whatever is considered as being independent, or a cause, or simple, universal, one, equal, like, straight, and so forth; and the absolute I call the simplest and the easiest of all, so that we can make use of it in the solution of questions.

But the relative is that which, while participating in the same nature, or at least sharing in it to some degree which enables us to relate it to the absolute and to deduce it from that by a chain of operations, involves in addition something else in its concept which I call relativity. Examples of this are found in whatever is said to be dependent, or an effect, composite, particular, many, unequal, unlike, oblique, etc. These relatives are the further removed from the absolute, in proportion as they contain more elements of relativity subordinate the one to the other. We state in this rule that these should all be distinguished and their correlative connection and natural order so observed, that we may be able by traversing all the intermediate steps to proceed from the most remote to that which is in the highest degree absolute.

Herein lies the secret of this whole method, that in all things we should diligently mark that which is most absolute. For some things are from one point of view more absolute than others, but from a different standpoint are more relative. Thus though the universal is more absolute than the particular because its essence is simpler, yet it can be held to be more relative than the latter, because it depends upon individuals for its existence, and so on. Certain things likewise are truly more absolute than others, but yet are not the most absolute of all. Thus relatively to individuals, species is something absolute, but contrasted with genus it is relative. So too, among things that can be measured, extension is something absolute, but among the various aspects of extension it is length that is absolute, and so on. Finally also, in order to bring out more clearly that we are considering here not the nature of each thing taken in isolation, but the series involved in knowing them, we have purposely enumerated cause and equality among our absolutes, though the nature of these terms is really relative. For though Philosophers make cause and effect correlative, we find that here even, if we ask what the effect is, we must first know the cause and not conversely. Equals too mutually imply one another, but we can know unequals only by comparing them with equals and not *per contra*.

Secondly, we must note that there are but few pure and simple essences, which either our experiences or some sort of light innate in us

enable us to behold as primary and existing *per se*, not as depending on any others. These we say should be carefully noticed, for they are just those facts which we have called the simplest in any single series. All the others can only be perceived as deductions from these, either immediate and proximate, or not to be attained save by two or three or more acts of inference. The number of these acts should be noted in order that we may perceive whether the facts are separated from the primary and simplest proposition by a greater or smaller number of steps. And so pronounced is everywhere the inter-connection of ground and consequence, which gives rise, in the objects to be examined, to those series to which every inquiry must be reduced, that it can be investigated by a sure method. But because it is not easy to make a review of them all, and besides, since they have not so much to be kept in the memory as to be detected by a sort of mental penetration, we must seek for something which will so mould our intelligence as to let it perceive these connected sequences immediately whenever it needs to do so. For this purpose I have found nothing so effectual as to accustom ourselves to turn our attention with a sort of penetrative insight on the very minutest of the facts which we have already discovered.

Finally, we must in the third place note that our inquiry ought not to start with the investigation of difficult matters. Rather, before setting out to attack any definite problem, it behoves us first, without making any selection, to assemble those truths that are obvious as they present themselves to us, and afterwards, proceeding step by step, to inquire whether any others can be deduced from these, and again any others from these conclusions and so on, in order. This done, we should attentively think over the truths we have discovered and mark with diligence the reasons why we have been able to detect some more easily than others, and which these are. Thus, when we come to attack some definite problem we shall be able to judge what previous questions it were best to settle first. For example, if it comes into my thought that the number 6 is twice 3, I may then ask what is twice 6, viz. 12; again, perhaps I seek for the double of this, viz. 24, and again of this, viz. 48. Thus I may easily deduce that there is the same proportion between 3 and 6, as between 6 and 12, and likewise 12 and 24, and so on, and hence that the numbers 3, 6, 12, 24, 48, etc. are in continued proportion. But though these facts are all so clear as to seem almost childish, I am now able by attentive reflection to understand what is the form involved by all questions that can be propounded about the proportions or relations of things. and the order in which they should be investigated; and this discovery embraces the sum of the entire science of Pure Mathematics.

For first I perceive that it was not more difficult to discover the double of six than that of three; and that equally in all cases, when we have found a proportion between any two magnitudes, we can find innumerable others which have the same proportion between them. So, too, there is no increase of difficulty, if three, or four, or more of such magnitudes are sought for, because each has to be found separately and without any relation to the others. But next I notice that though, when the magnitudes 3 and 6 are given, one can easily find a third in continued proportion, viz. 12, it is yet not equally easy, when the two extremes, 3 and 12, are given, to find the mean proportional, viz. 6. When we look into the reason for this, it is clear that here we have a type of difficulty quite different from the former; for, in order to find the mean proportional, we must at the same time attend to the two extremes and to the proportion which exists between these two in order to discover a new ratio by dividing the previous one; and this is a very different thing from finding a third term in continued proportion with two given numbers. I go forward likewise and examine whether, when the numbers 3 and 24 were given, it would have been equally easy to determine one of the two intermediate proportionals, viz. 6 and 12. But here still another sort of difficulty arises more involved than the previous ones, for on this occasion we have to attend not to one or two things only but to three, in order to discover the fourth. We may go still further and inquire whether if only 3 and 48 had been given it would have been still more difficult to discover one of the three mean proportionals, viz. 6, 12, and 24. At the first blush this indeed appears to be difficult; but immediately afterwards it comes to mind that this difficulty can be split up and lessened, if first of all we ask only for the mean proportional between 3 and 48, viz. 12, and then seek for the other mean proportional between 3 and 12, viz. 6, and the other between 12 and 48, viz. 24. Thus, we have reduced the problem to the difficulty of the second type shown above.

These illustrations further lead me to note

that the quest for knowledge about the same thing can traverse different routes, the one much more difficult and obscure than the other. Thus, to find these four continued proportionals, 3, 6, 12, and 24, if two consecutive numbers be assumed, e.g. 3 and 6, or 6 and 12, or 12 and 24, in order that we may discover the others, our task will be easy. In this case we shall say that the proposition to be discovered is directly examined. But if the two numbers given are alternates, like 3 and 12, or 6 and 24, which are to lead us to the discovery of the others, then we shall call this an indirect investigation of the first mode. Likewise, if we are given two extremes like 3 and 24, in order to find out from these the intermediates 6 and 12, the investigation will be indirect and of the second mode. Thus I should be able to proceed further and deduce many other results from this example; but these will be sufficient, if the reader follows my meaning when I say that a proposition is directly deduced, or indirectly, and will reflect that from a knowledge of each of these matters that are simplest and primary, much may be discovered in other sciences by those who bring to them attentive thought and a power of sagacious analysis.

RULE VII

If we wish our science to be complete, those matters which promote the end we have in view must one and all be scrutinized by a movement of thought which is continuous and nowhere interrupted; they must also be included in an enumeration which is both adequate and methodical.

It is necessary to obey the injunctions of this rule if we hope to gain admission among the certain truths for those which, we have declared above, are not immediate deductions from primary and self-evident principles. For this deduction frequently involves such a long series of transitions from ground to consequent that when we come to the conclusion we have difficulty in recalling the whole of the route by which we have arrived at it. This is why I say that there must be a continuous movement of thought to make good this weakness of the memory. Thus, e.g. if I have first found out by separate mental operations what the relation is between the magnitudes A and B, then what between B and C, between C and D, and finally between D and E, that does not entail my seeing what the relation is between A and E, nor can the truths previously learnt give me a precise knowledge of it unless I recall them all. To remedy this I would run them over from time to time, keeping the imagination moving continuously in such a way that while it is intuitively perceiving each fact it simultaneously passes on to the next; and this I would do until I had learned to pass from the first to the last so quickly, that no stage in the process was left to the care of the memory, but I seemed to have the whole in intuition before me at the same time. This method will both relieve the memory, diminish the sluggishness of our thinking, and definitely enlarge our mental capacity.

But we must add that this movement should nowhere be interrupted. Often people who attempt to deduce a conclusion too quickly and from remote principles do not trace the whole chain of intermediate conclusions with sufficient accuracy to prevent them from passing over many steps without due consideration. But it is certain that wherever the smallest link is left out the chain is broken and the whole of the certainty of the conclusion falls to the ground.

Here we maintain that an enumeration [of the steps in a proof] is required as well, if we wish to make our science complete. For resolving most problems other precepts are profitable, but enumeration alone will secure our always passing a true and certain judgment on whatsoever engages our attention; by means of it nothing at all will escape us, but we shall evidently have some knowledge of every step.

This enumeration or induction is thus a review or inventory of all those matters that have a bearing on the problem raised, which is so thorough and accurate that by its means we can clearly and with confidence conclude that we have omitted nothing by mistake. Consequently as often as we have employed it, if the problem defies us, we shall at least be wiser in this respect, viz. that we are quite certain that we know of no way of resolving it. If it chances, as often it does, that we have been able to scan all the routes leading to it which lie open to the human intelligence, we shall be entitled boldly to assert that the solution of the problem lies outside the reach of human knowledge.

Furthermore, we must note that by adequate enumeration or induction is only meant that method by which we may attain surer conclusions than by any other type of proof, with the exception of simple intuition. But when the knowledge of some matter cannot be reduced to this, we must cast aside all syllogistic fetters and employ induction, the only method

left us, but one in which all confidence should be reposed. For whenever single facts have been immediately deduced the one from the other, they have been already reduced, if the inference was evident, to a true intuition. But if we infer any single thing from various and disconnected facts, often our intellectual capacity is not so great as to be able to embrace them all in a single intuition; in which case our mind should be content with the certitude attaching to this operation. It is in precisely similar fashion that though we cannot with one single gaze distinguish all the links of a lengthy chain, yet if we have seen the connection of each with its neighbour, we shall be entitled to say that we have seen how the first is connected with the last.

I have declared that this operation ought to be adequate because it is often in danger of being defective and consequently exposed to error. For sometimes, even though in our enumeration we scrutinize many facts which are highly evident, yet if we omit the smallest step the chain is broken and the whole of the certitude of the conclusion falls to the ground. Sometimes also, even though all the facts are included in an accurate enumeration, the single steps are not distinguished from one another, and our knowledge of them all is thus only confused.

Further, while now the enumeration ought to be complete, now distinct, there are times when it need have neither of these characters; it was for this reason that I said only that it should be adequate. For if I want to prove by enumeration how many genera there are of corporeal things, or of those that in any way fall under the senses, I shall not assert that they are just so many and no more, unless I previously have become aware that I have included them all in my enumeration, and have distinguished them each separately from all the others. But if in the same way I wish to prove that the rational soul is not corporeal, I do not need a complete enumeration; it will be sufficient to include all bodies in certain collections in such a way as to be able to demonstrate that the rational soul has nothing to do with any of these. If, finally, I wish to show by enumeration that the area of a circle is greater than the area of all other figures whose perimeter is equal, there is no need for me to call in review all other figures; it is enough to demonstrate this of certain others in particular, in order to get thence by induction the same conclusion about all the others.

I added also that the enumeration ought to be methodical. This is both because we have no more serviceable remedy for the defects already instanced, than to scan all things in an orderly manner; and also because it often happens that if each single matter which concerns the quest in hand were to be investigated separately, no man's life would be long enough for the purpose, whether because they are far too many, or because it would chance that the same things had to be repeated too often. But if all these facts are arranged in the best order, they will for the most part be reduced to determinate classes, out of which it will be sufficient to take one example for exact inspection, or some one feature in a single case, or certain things rather than others, or at least we shall never have to waste our time in traversing the same ground twice. The advantage of this course is so great that often many particulars can, owing to a well devised arrangement, be gone over in a short space of time and with little trouble, though at first view the matter looked immense.

But this order which we employ in our enumerations can for the most part be varied and depends upon each man's judgment. For this reason, if we would elaborate it in our thought with greater penetration, we must remember what was said in our fifth proposition. There are also many of the trivial things of man's devising, in the discovery of which the whole method lies in the disposal of this order. Thus if you wish to construct a perfect anagram by the transposition of the letters of a name, there is no need to pass from the easy to the difficult, nor to distinguish absolute from relative. Here there is no place for these operations; it will be sufficient to adopt an order to be followed in the transpositions of the letters which we are to examine, such that the same arrangements are never handled twice over. The total number of transpositions should, e.g. be split up into definite classes, so that it may immediately appear in which there is the best hope of finding what is sought. In this way the task is often not tedious but merely child's play.

However, these three propositions should not be separated, because for the most part we have to think of them together, and all equally tend towards the perfecting of our method. There was no great reason for treating one before the other, and we have expounded them but briefly here. The reason for this is that in the rest of the treatise we have practically nothing else left for consideration. Therefore,

we shall then exhibit in detail what here we have brought together in a general way.

RULE VIII

If in the matters to be examined we come to a step in the series of which our understanding is not sufficiently well able to have an intuitive cognition, we must stop short there. We must make no attempt to examine what follows; thus we shall spare ourselves superfluous labour.

THE THREE preceding rules prescribe and explain the order to be followed. The present rule, on the other hand, shows when it is wholly necessary and when it is merely useful. Thus it is necessary to examine whatever constitutes a single step in that series, by which we pass from relative to absolute, or conversely, before discussing what follows from it. But if, as often happens, many things pertain to the same step, though it is indeed always profitable to review them in order, in this case we are not forced to apply our method of observation so strictly and rigidly. Frequently it is permissible to proceed farther, even though we have not clear knowledge of all the facts it involves, but know only a few or a single one of them.

This rule is a necessary consequence of the reasons brought forward in support of the second. But it must not be thought that the present rule contributes nothing fresh towards the advancement of learning, though it seems only to bid us refrain from further discussion, and apparently does not unfold any truth. For beginners, indeed, it has no further value than to teach them how not to waste time, and it employs nearly the same arguments in doing so as Rule II. But it shows those who have perfectly mastered the seven preceding maxims, how in the pursuit of any science so to satisfy themselves as not to desire anything further. For the man who faithfully complies with the former rules in the solution of any difficulty, and yet by the present rule is bidden desist at a certain point, will then know for certainty that no amount of application will enable him to attain to the knowledge desired, and that not owing to a defect in his intelligence, but because the nature of the problem itself, or the fact that he is human, prevents him. But this knowledge is not the less science than that which reveals the nature of the thing itself; in fact he would seem to have some mental defect who should extend his curiosity farther.

But what we have been saying must be illustrated by one or two examples. If, for example, one who studies only Mathematics were to seek to find that curve which in dioptrics is called the anaclastic, that from which parallel rays are so refracted that after the refraction they all meet in one point,—it will be easy to see, by applying Rules V and VI, that the determination of this line depends upon the relation which the angles of refraction bear to the angles of incidence. But because he is unable to discover this, since it is a matter not of Mathematics but of Physics, he is here forced to pause at the threshold. Nor will it avail him to try and learn this from the Philosophers or to gather it from experience; for this would be to break Rule III. Furthermore, this proposition is both composite and relative; but in the proper place we shall show that experience is unambiguous only when dealing with the wholly simple and absolute. Again, it will be vain for him to assume some relation or other as being that which prevails between such angles, and conjecture that this is the truest to fact; for in that case he would be on the track not of the anaclastic, but merely of that curve which could be deduced from his assumption.

If, however, a man who does not confine his studies to Mathematics, but, in accordance with the first rule, tries to discover the truth on all points, meets with the same difficulty, he will find in addition that this ratio between the angles of incidence and of refraction depends upon changes in their relation produced by varying the medium. Again these changes depend upon the manner in which the ray of light traverses the whole transparent body; while the knowledge of the way in which the light thus passes through presupposes a knowledge of the nature of the action of light, to understand which finally we must know what a natural potency is in general, this last being the most absolute term in the whole series in question. When, therefore, by a mental intuition he has clearly comprehended the nature of this, he will, in compliance with Rule V, proceed backwards by the same steps. And if when he comes to the second step he is unable straightway to determine the nature of light, he will, in accordance with the seventh rule enumerate all the other natural potencies, in order that the knowledge of some other of them may help him, at least by analogy (of which more anon), to understand this. This done, he will ask how the ray traverses the whole of the transparent body, and will so follow out the other points methodically, that at last he will arrive at the anaclastic itself. Though this has long defied the efforts of many

inquirers, I see no reason why a man who fully carried out our method should fail to arrive at a convincing knowledge of the matter.

But let us give the most splendid example of all. If a man proposes to himself the problem of examining all the truths for the knowledge of which human reason suffices—and I think that this is a task which should be undertaken once at least in his life by every person who seriously endeavours to attain equilibrium of thought—, he will, by the rules given above, certainly discover that nothing can be known prior to the understanding, since the knowledge of all things else depends upon this and not conversely. Then, when he has clearly grasped all those things which follow proximately on the knowledge of the naked understanding, he will enumerate among other things whatever instruments of thought we have other than the understanding; and these are only two, viz. imagination and sense. He will therefore devote all his energies to the distinguishing and examining of these three modes of cognition, and seeing that in the strict sense truth and falsity can be a matter of the understanding alone, though often it derives its origin from the other two faculties, he will attend carefully to every source of deception in order that he may be on his guard. He will also enumerate exactly all the ways leading to truth which lie open to us, in order that he may follow the right way. They are not so many that they cannot all be easily discovered and embraced in an adequate enumeration. And though this will seem marvellous and incredible to the inexpert, as soon as in each matter he has distinguished those cognitions which only fill and embellish the memory, from those which cause one to be deemed really more instructed, which it will be easy for him to do . . . ; he will feel assured that any absence of further knowledge is not due to lack of intelligence or of skill, and that nothing at all can be known by anyone else which he is not capable of knowing, provided only that he gives to it his utmost mental application. And though many problems may present themselves, from the solution of which this rule prohibits him, yet because he will clearly perceive that they pass the limits of human intelligence, he will deem that he is not the more ignorant on that account; rather, if he is reasonable, this very knowledge, that the solution can be discovered by no one, will abundantly satisfy his curiosity.

But lest we should always be uncertain as to the powers of the mind, and in order that w may not labour wrongly and at random befo we set ourselves to think out things in detai we ought once in our life to inquire diligentl what the thoughts are of which the huma mind is capable. In order the better to attai this end we ought, when two sets of inquirie are equally simple, to choose the more usefu

This method of ours resembles indeed thos devices employed by the mechanical craft which do not need the aid of anything outsid of them, but themselves supply the directio for making their own instruments. Thus if man wished to practise any one of them, e.g the craft of a smith, and were destitute of a instruments, he would be forced to use at firs a hard stone or a rough lump of iron as an an vil, take a piece of rock in place of a hamme make pieces of wood serve as tongs, and pr vide himself with other such tools as necessit required. Thus equipped, he would not then a once attempt to forge swords or helmets or an manufactured article of iron for others to us He would first of all fashion hammer, anvi tongs, and the other tools useful for himsel This example teaches us that, since thus at th outset we have been able to discover only som rough precepts, apparently the innate posse sion of our mind, rather than the product c technical skill, we should not forthwith a tempt to settle the controversies of Philos phers, or solve the puzzles of the Mathemat cians, by their help. We must first emplo them for searching out with our utmost atte tion all the other things that are more urgentl required in the investigation of truth. And th since there is no reason why it should appea more difficult to discover these than any of th answers which the problems propounded b Geometry or Physics or the other sciences an wont to demand.

Now no more useful inquiry can be propose than that which seeks to determine the natu and the scope of human knowledge. This why we state this very problem succinctly i the single question, which we deem should b answered at the very outset with the aid of th rules which we have already laid down. Th investigation should be undertaken once a least in his life by anyone who has the slighte regard for truth, since in pursuing it the tru instruments of knowledge and the whole met od of inquiry come to light. But nothi seems to me more futile than the conduct o those who boldly dispute about the secrets o nature, the influence of the heavens on the

lower regions, the predicting of future events and similar matters, as many do, without yet having ever asked even whether human reason is adequate to the solution of these problems. Neither ought it to seem such a toilsome and difficult matter to define the limits of that understanding of which we are directly aware as being with us, when we often have no hesitation in passing judgment even on things that are without us and quite foreign to us. Neither is it such an immense task to attempt to grasp in thought all the objects comprised within this whole of things, in order to discover how they singly fall under our mental scrutiny. For nothing can prove to be so complex or so vague as to defeat the efforts of the method of enumeration above described, directed towards restraining it within certain limits or arranging it under certain categories. But to put this to the test in the matter of the question above propounded, we first of all divide the whole problem relative to it into two parts; for it ought either to relate to us who are capable of knowledge, or to the things themselves which can be known: and these two factors we discuss separately.

In ourselves we notice that while it is the understanding alone which is capable of knowing, it yet is either helped or hindered by three other faculties, namely imagination, sense and memory. We must therefore examine these faculties in order, with a view to finding out where each may prove to be an impediment, so that we may be on our guard; or where it may profit us, so that we may use to the full the resources of these powers. This first part of our problem will accordingly be discussed with the aid of a sufficient enumeration, as will be shown in the succeeding proposition.

We come secondly to the things themselves which must be considered only in so far as they are the objects of the understanding. From this point of view we divide them into the class (1) of those whose nature is of the extremest simplicity and (2) of the complex and composite. Simple natures must be either spiritual or corporeal or at once spiritual and corporeal. Finally, among the composites there are some which the understanding realises to be complex before it judges that it can determine anything about them; but there are also others which it itself puts together. All these matters will be expounded at greater length in the twelfth proposition, where it will be shown that there can be no falsity save in the last class—that of the compounds made by the understanding itself. This is why we further subdivide these into the class of those which are deducible from natures which are of the maximum simplicity and are known *per se*, of which we shall treat in the whole of the succeeding book[1]; and into those which presuppose the existence of others which the facts themselves show us to be composite. To the exposition of these we destine the whole of the third[2] book.

But we shall indeed attempt in the whole of this treatise to follow so accurately the paths which conduct men to the knowledge of the truth, and to make them so easy, that anyone who has perfectly learned the whole of this method, however moderate may be his talent, may see that no avenue to the truth is closed to him from which everyone else is not also excluded, and that his ignorance is due neither to a deficiency in his capacity nor to his method of procedure. But as often as he applies his mind to the understanding of some matter, he will either be entirely successful, or he will realise that success depends upon a certain experiment which he is unable to perform, and in that case he will not blame his mental capacity although he is compelled to stop short there. Or finally, he may show that the knowledge desired wholly exceeds the limits of the human intelligence; and consequently he will believe that he is none the more ignorant on that account. For to have discovered this is knowledge in no less degree than the knowledge of anything else.

RULE IX

We ought to give the whole of our attention to the most insignificant and most easily mastered facts, and remain a long time in contemplation of them until we are accustomed to behold the truth clearly and distinctly.

We have now indicated the two operations of our understanding, intuition and deduction, on which alone we have said we must rely in the acquisition of knowledge. Let us therefore in this and in the following proposition proceed to explain how we can render ourselves more skilful in employing them, and at the same time cultivate the two principal faculties of the mind, to wit perspicacity, by viewing single objects distinctly, and sagacity, by the skilful deduction of certain facts from others.

[1]This begins at Rule XIII. Of the later rules we have the titles only in the case of XIX—XXI, while the last three are entirely lacking

[2]Apparently not even begun.

Truly we shall learn how to employ our mental intuition from comparing it with the way in which we employ our eyes. For he who attempts to view a multitude of objects with one and the same glance, sees none of them distinctly; and similarly the man who is wont to attend to many things at the same time by means of a single act of thought is confused in mind. But just as workmen, who are employed in very fine and delicate operations and are accustomed to direct their eyesight attentively to separate points, by practice have acquired a capacity for distinguishing objects of extreme minuteness and subtlety; so likewise people, who do not allow their thought to be distracted by various objects at the same time, but always concentrate it in attending to the simplest and easiest particulars, are clear-headed.

But it is a common failing of mortals to deem the more difficult the fairer; and they often think that they have learned nothing when they see a very clear and simple cause for a fact, while at the same time they are lost in admiration of certain sublime and profound philosophical explanations, even though these for the most part are based upon foundations which no one had adequately surveyed—a mental disorder which prizes the darkness higher than the light. But it is notable that those who have real knowledge discern the truth with equal facility whether they evolve it from matter that is simple or that is obscure; they grasp each fact by an act of thought that is similar, single, and distinct, after they have once arrived at the point in question. The whole of the difference between the apprehension of the simple and of the obscure lies in the route taken, which certainly ought to be longer if it conducts us from our initial and most absolute principles to a truth that is somewhat remote.

Everyone ought therefore to accustom himself to grasp in his thought at the same time facts that are at once so few and so simple, that he shall never believe that he has knowledge of anything which he does not mentally behold with a distinctness equal to that of the objects which he knows most distinctly of all. It is true that some men are born with a much greater aptitude for such discernment than others, but the mind can be made much more expert at such work by art and exercise. But there is one fact which I should here emphasize above all others; and that is that everyone should firmly persuade himself that none of the sciences, however abstruse, is to be deduced from lofty and obscure matters, but that they all proceed only from what is easy and more readily understood.

For example if I wish to examine whether it is possible for a natural force to pass at one and the same moment to a spot at a distance and yet to traverse the whole space in between, I shall not begin to study the force of magnetism or the influence of the stars, not even the speed of light, in order to discover whether actions such as these occur instantaneously; for the solution of this question would be more difficult than the problem proposed. I should rather bethink myself of the spatial motions of bodies, because nothing in the sphere of motion can be found more obvious to sense than this. I shall observe that while a stone cannot pass to another place in one and the same moment, because it is a body, yet a force similar to that which moves the stone is communicated exactly instantaneously if it passes unencumbered from one object to another. For instance, if I move one end of a stick of whatever length, I easily understand that the power by which that part of the stick is moved necessarily moves also all its other parts at the same moment, because then the force passes unencumbered and is not imprisoned in any body, e.g. a stone, which bears it along.

In the same way if I wish to understand how one and the same simple cause can produce contrary effects at the same time, I shall not cite the drugs of the doctors which expel certain humours and retain others; nor shall I romance about the moon's power of warming with its light and chilling by means of some occult power. I shall rather cast my eyes upon the balance in which the same weight raises one arm at the same time as it depresses the other, or take some other familiar instance.

RULE X

In order that it may acquire sagacity the mind should be exercised in pursuing just those inquiries of which the solution has already been found by others; and it ought to traverse in a systematic way even the most trifling of men's inventions though those ought to be preferred in which order is explained or implied.

I CONFESS that my natural disposition is such that I have always found, not the following of the arguments of others, but the discovery of reasons by my own proper efforts, to yield me the highest intellectual satisfaction. It was this alone that attracted me, when I was still a

young man, to the study of science. And whenever any book by its title promised some new discovery, before I read further I tried whether I could achieve something similar by means of some inborn faculty of invention, and I was careful lest a premature perusal of the book might deprive me of this harmless pleasure. So often was I successful that at length I perceived that I no longer came upon the truth by proceeding as others commonly do, viz. by pursuing vague and blind inquiries and relying more on good fortune than on skill. I saw that by long experience I had discovered certain rules which are of no little help in this inquiry, and which I used afterwards in devising further rules. Thus it was that I diligently elaborated the whole of this method and came to the conclusion that I had followed that plan of study which was the most fruitful of all.

But because not all minds are so much inclined to puzzle things out unaided, this proposition announces that we ought not immediately to occupy ourselves with the more difficult and arduous problems, but first should discuss those disciplines which are easiest and simplest, and those above all in which order most prevails. Such are the arts of the craftsmen who weave webs and tapestry, or of women who embroider or use in the same work threads with infinite modification of texture. With these are ranked all play with numbers and everything that belongs to Arithmetic, and the like. It is wonderful how all these studies discipline our mental powers, provided that we do not know the solutions from others, but invent them ourselves. For since nothing in these arts remains hidden, and they are wholly adjusted to the capacity of human cognition, they reveal to us with the greatest distinctness innumerable orderly systems, all different from each other, but none the less conforming to rule, in the proper observance of which systems of order consists the whole of human sagacity.

It was for this reason that we insisted that method must be employed in studying these matters; and this in those arts of less importance consists wholly in the close observation of the order which is found in the object studied, whether that be an order existing in the thing itself, or due to subtle human devising. Thus if we wish to make out some writing in which the meaning is disguised by the use of a cypher, though the order here fails to present itself, we yet make up an imaginary one, for the purpose both of testing all the conjectures we may make about single letters, words or sentences, and in order to arrange them so that when we sum them up we shall be able to tell all the inferences that we can deduce from them. We must principally beware of wasting our time in such cases by proceeding at random and unmethodically; for even though the solution can often be found without method, and by lucky people sometimes quicker, yet such procedure is likely to enfeeble the faculties and to make people accustomed to the trifling and the childish, so that for the future their minds will stick on the surface of things, incapable of penetrating beyond it. But meanwhile we must not fall into the error of those who, having devoted themselves solely to what is lofty and serious, find that after many years of toil they have acquired, not the profound knowledge they hoped for, but only mental confusion. Hence we must give ourselves practice first in those easier disciplines, but methodically, so that by open and familiar ways we may ceaselessly accustom ourselves to penetrate as easily as though we were at play into the very heart of these subjects. For by this means we shall afterwards gradually feel (and in a space of time shorter than we could at all hope for) that we are in a position with equal facility to deduce from evident first principles many propositions which at first sight are highly intricate and difficult.

It may perhaps strike some with surprise that here, where we are discussing how to improve our power of deducing one truth from another, we have omitted all the precepts of the dialecticians, by which they think to control the human reason. They prescribe certain formulae of argument, which lead to a conclusion with such necessity that, if the reason commits itself to their trust, even though it slackens its interest and no longer pays a heedful and close attention to the very proposition inferred, it can nevertheless at the same time come to a sure conclusion by virtue of the form of the argument alone. Exactly so; the fact is that frequently we notice that often the truth escapes away out of these imprisoning bonds, while the people themselves who have used them in order to capture it remain entangled in them. Other people are not so frequently entrapped; and it is a matter of experience that the most ingenious sophisms hardly ever impose on anyone who uses his unaided reason, while they are wont to deceive the sophists themselves.

Wherefore as we wish here to be particularly careful lest our reason should go on holiday while we are examining the truth of any matter, we reject those formulae as being opposed to our project, and look out rather for all the aids by which our thought may be kept attentive, as will be shown in the sequel. But, to say a few words more, that it may appear still more evident that this style of argument contributes nothing at all to the discovery of the truth, we must note that the Dialecticians are unable to devise any syllogism which has a true conclusion, unless they have first secured the material out of which to construct it, i.e. unless they have already ascertained the very truth which is deduced in that syllogism. Whence it is clear that from a formula of this kind they can gather nothing that is new, and hence the ordinary Dialectic is quite valueless for those who desire to investigate the truth of things. Its only possible use is to serve to explain at times more easily to others the truths we have already ascertained; hence it should be transferred from Philosophy to Rhetoric.

RULE XI

If, after we have recognized intuitively a number of simple truths, we wish to draw any inference from them, it is useful to run them over in a continuous and uninterrupted act of thought, to reflect upon their relations to one another, and to grasp together distinctly a number of these propositions so far as is possible at the same time. For this is a way of making our knowledge much more certain, and of greatly increasing the power of the mind.

HERE we have an opportunity of expounding more clearly what has been already said of mental intuition in the third and seventh rules. In one passage[1] we opposed it to deduction, while in the other we distinguished it from enumeration only, which we defined as an inference drawn from many and diverse things[2]. But the simple deduction of one thing from another, we said in the same passage[3], was effected by intuition.

It was necessary to do this, because two things are requisite for mental intuition. Firstly, the proposition intuited must be clear and distinct; secondly, it must be grasped in its totality at the same time and not successively. As for deduction, if we are thinking of how the process works, as we were in Rule III, it appears not to occur all at the same time, but involves a sort of movement on the part of our mind when it infers one thing from another. We were justified therefore in distinguishing deduction in that rule from intuition. But if we wish to consider deduction as an accomplished fact, as we did in what we said relatively to the seventh rule, then it no longer designates a movement, but rather the completion of a movement, and therefore we suppose that it is presented to us by intuition when it is simple and clear, but not when it is complex and involved. When this is the case we give it the name of enumeration or induction, because it cannot then be grasped as a whole at the same time by the mind, and its certainty depends to some extent on the memory, in which our judgments about the various matters enumerated must be retained, if from their assemblage a single fact is to be inferred.

All these distinctions had to be made if we were to elucidate this rule. We treated of mental intuition solely in Rule IX; the tenth dealt with enumeration alone; but now the present rule explains how these two operations aid and complete each other. In doing so they seem to grow into a single process by virtue of a sort of motion of thought which has an attentive and vision-like knowledge of one fact and yet can pass at the very same moment to another.

Now to this co-operation we assign a two-fold advantage. Firstly, it promotes a more certain knowledge of the conclusion with which we are concerned, and secondly, it makes the mind readier to discover fresh truths. In fact the memory, on which we have said depends the certainty of the conclusions which embrace more than we can grasp in a single act of intuition, though weak and liable to fail us, can be renewed and made stronger by this continuous and constantly repeated process of thought. Thus if diverse mental acts have led me to know what is the relation between a first and a second magnitude, next between the second and a third, then between the third and a fourth, and finally the fourth and a fifth, that need not lead me to see what is the relation between the first and the fifth, nor can I deduce it from what I already know, unless I remember all the other relations. Hence what I have to do is to run over them all repeatedly in my mind, until I pass so quickly from the first to the last that practically no step is left to the memory, and I seem to view the whole all at the same time.

Everyone must see that this plan does much

[1] Cf. p. 4.
[2] Cf. p. 10.
[3] Cf. p. 4.

to counteract the slowness of the mind and to enlarge its capacity. But in addition we must note that the greatest advantage of this rule consists in the fact that, by reflecting on the mutual dependence of two propositions, we acquire the habit of distinguishing at a glance what is more or less relative, and what the steps are by which a relative fact is related to something absolute. For example, if I run over a number of magnitudes that are in continued proportion, I shall reflect upon all the following facts: viz. that the mental act is entirely similar—and not easier in the one case, more difficult in another—by which I grasp the relation between the first and the second, the second and third, third and fourth, and so on; while yet it is more difficult for me to conceive what the relation of the second is to the first and to the third at the same time, and much more difficult still to tell its relation to the first and fourth, and so on. These considerations then lead me to see why, if the first and second alone are given, I can easily find the third and fourth, and all the others; the reason being that this process requires only single and distinct acts of thought. But if only the first and the third are given, it is not so easy to recognize the mean, because this can only be accomplished by means of a mental operation in which two of the previous acts are involved. If the first and the fourth magnitudes alone are given, it is still more difficult to present to ourselves the two means, because here three acts of thought come in simultaneously. It would seem likely as a consequence that it would be even more difficult to discover the three means between the first and the fifth. The reason why this is not so is due to a fresh fact; viz. even though here four mental acts come together they can yet be disjoined, since four can be divided by another number. Thus I can discover the third by itself from the first and fifth, then the second from the first and third, and so on. If one accustoms one's self to reflect on these and similar problems, as often as a new question arises, at once one recognizes what produces its special difficulty, and what is the simplest method of dealing with all cases; and to be able to do so is a valuable aid to the discovery of the truth.

RULE XII

Finally we ought to employ all the aids of understanding, imagination, sense and memory, first for the purpose of having a distinct intuition of simple propositions; partly also in order to compare the propositions to be proved with those we know already, so that we may be able to recognize their truth; partly also in order to discover the truths, which should be compared with each other so that nothing may be left lacking on which human industry may exercise itself.

This rule states the conclusion of all that we said before, and shows in general outline what had to be explained in detail, in this wise.

In the matter of cognition of facts two things alone have to be considered, ourselves who know and the objects themselves which are to be known. Within us there are four faculties only which we can use for this purpose, viz. understanding, imagination, sense and memory. The understanding is indeed alone capable of perceiving the truth, but yet it ought to be aided by imagination, sense and memory, lest perchance we omit any expedient that lies within our power. On the side of the facts to be known it is enough to examine three things; first, that which presents itself spontaneously, secondly, how we learn one thing by means of another, and thirdly, what (truths) are deduced from what. This enumeration appears to me to be complete, and to omit nothing to which our human powers can apply.

I should have liked therefore to have turned to the first point and to have explained in this passage, what the human mind is, what body, and how it is "informed" by mind; what the faculties in the complex whole are which serve the attainment of knowledge, and what the agency of each is. But this place seems hardly to give me sufficient room to take in all the matters which must be premised before the truth in this subject can become clear to all. For my desire is in all that I write to assert nothing controversial unless I have already stated the very reasons which have brought me to that conclusion, and by which I think that others also may be convinced.

But because at present I am prevented from doing this, it will suffice me to explain as briefly as possible that mode of viewing everything within us which is directed towards the discovery of truth, which most promotes my purpose. You need not believe that the facts are so unless you like. But what prevents us following these suppositions, if it appears that they do no harm to the truth, but only render it all much clearer? In Geometry you do precisely the same thing when you make certain assumptions about a quantity which do not in any way weaken the force of your arguments, though often our experience of its nature in

Physics makes us judge of it quite otherwise.

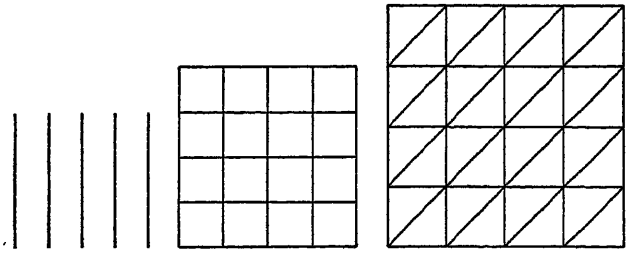
Let us then conceive of the matter as follows:—all our external senses, in so far as they are part of the body, and despite the fact that we direct them towards objects, so manifesting activity, viz. a movement in space, nevertheless properly speaking perceive in virtue of passivity alone, just in the way that wax receives an impression from a seal. And it should not be thought that all we mean to assert is an analogy between the two. We ought to believe that the way is entirely the same in which the exterior figure of the sentient body is really modified by the object, as that in which the shape of the surface of the wax is altered by the seal. This has to be admitted not only in the case of the figure, hardness, roughness, etc. of a body which we perceive by touch, but even when we are aware of heat, cold, and the like qualities. It is likewise with the other senses. The first opaque structure in the eye receives the figure impressed upon it by the light with its various colours; and the first membrane in the ears, the nose, and the tongue that resists the further passage of the object, thus also acquires a new figure from the sound, the odour, and the savour, as the case may be.

It is exceedingly helpful to conceive all those matters thus, for nothing falls more readily under sense than figure, which can be touched and seen. Moreover that nothing false issues from this supposition more than from any other, is proved by the fact that the concept of figure is so common and simple that it is involved in every object of sense. Thus whatever you suppose colour to be, you cannot deny that it is extended and in consequence possessed of figure. Is there then any disadvantage, if, while taking care not to admit any new entity uselessly, or rashly to imagine that it exists, and not denying indeed the beliefs of others concerning colour, but merely abstracting from every other feature except that it possesses the nature of figure, we conceive the diversity existing between white, blue, and red, etc., as being like the difference between the following similar figures? The same argument applies to all cases; for it is certain that the infinitude of figures suffices to express all the differences in sensible things.

Secondly, we must believe that while the external sense is stimulated by the object, the figure which is conveyed to it is carried off to some other part of the body, that part called the common sense, in the very same instant and without the passage of any real entity from one to the other. It is in exactly the same manner that now when I write I recognize that at the very moment when the separate characters are being written down on the paper, not only is the lower end of the pen moved, but every motion in that part is simultaneously shared by the whole pen. All these diverse motions are traced by the upper end of the pen likewise in the air, although I do not conceive of anything real passing from the one extremity to the other. Now who imagines that the connection between the different parts of the human body is slighter than that between the ends of a pen, and what simpler way of expressing this could be found?

Thirdly, we must believe that the common sense has a function like that of a seal, and impresses on the fancy or imagination, as though on wax, those very figures and ideas which come uncontaminated and without bodily admixture from the external senses. But this fancy is a genuine part of the body, of sufficient size to allow its different parts to assume various figures in distinctness from each other and to let those parts acquire the practice of retaining the impressions for some time. In the latter case we give the faculty the name of memory.

In the fourth place, we must conceive that the motor force or the nerves themselves derive their origin from the brain, in which the fancy is located, and that the fancy moves them in various ways, just as the external senses act on the common sense, or the lower extremity of the pen moves the whole pen. This example also shows how the fancy can be the cause of many motions in the nerves, motions of which, however, it does not have the images stamped upon it, possessing only certain other images from which these latter follow. Just so the whole pen does not move exactly in the way in which its lower end does; nay the greater part seems



to have a motion that is quite different from and contrary to that of the other. This lets us understand how all the motions of the other animals can come about, though we can ascribe to them no knowledge at all, but only fancy of a purely corporeal kind. We can explain also how in ourselves all those operations occur which we perform without any aid from the reason.

Finally and in the fifth place, we must think that that power by which we are properly said to know things, is purely spiritual, and not less distinct from every part of the body than blood from bone, or hand from eye. It is a single agency, whether it receives impressions from the common sense simultaneously with the fancy, or applies itself to those that are preserved in the memory, or forms new ones. Often the imagination is so beset by these impressions that it is unable at the same time to receive ideas from the common sense, or to transfer them to the motor mechanism in the way befitting its purely corporeal character. In all these operations this cognitive power is at one time passive, at another active, and resembles now the seal and now the wax. But the resemblance on this occasion is only one of analogy, for among corporeal things there is nothing wholly similar to this faculty. It is one and the same agency which, when applying itself along with the imagination to the common sense, is said to see, touch, etc.; if applying itself to the imagination alone in so far as that is endowed with diverse impressions, it is said to remember; if it turn to the imagination in order to create fresh impressions, it is said to imagine or conceive; finally if it act alone it is said to understand. How this latter function takes place I shall explain at greater length in the proper place. Now it is the same faculty that in correspondence with those various functions is called either pure understanding, or imagination, or memory, or sense. It is properly called mind when it either forms new ideas in the fancy, or attends to those already formed. We consider it as capable of the above various operations, and this distinction between those terms must in the sequel be borne in mind. But after having grasped these facts the attentive reader will gather what help is to be expected from each particular faculty, and discover how far human effort can avail to supplement the deficiencies of our mental powers.

For, since the understanding can be stimulated by the imagination, or on the contrary act on it; and seeing that the imagination can act on the senses by means of the motor power applying them to objects, while they on the contrary can act on it, depicting on it the images of bodies; considering on the other hand that the memory, at least that which is corporeal and similar to that of the brutes, is in no respect distinct from the imagination; we come to the sure conclusion that, if the understanding deal with matters in which there is nothing corporeal or similar to the corporeal, it cannot be helped by those faculties, but that, on the contrary, to prevent their hampering it, the senses must be banished and the imagination as far as possible divested of every distinct impression. But if the understanding proposes to examine something that can be referred to the body, we must form the idea of that thing as distinctly as possible in the imagination; and in order to effect this with greater ease, the thing itself which this idea is to represent must be exhibited to the external senses. Now when the understanding wishes to have a distinct intuition of particular facts a multitude of objects is of no use to it. But if it wishes to deduce one thing from a number of objects, as often has to be done, we must banish from the ideas of the objects presented whatsoever does not require present attention, in order that the remainder may be the more readily retained in memory. In the same way it is not on those occasions that the objects themselves ought to be presented to the external senses, but rather certain compendious abbreviations which, provided they guard the memory against lapse, are the handier the shorter they are. Whosoever observes all these recommendations, will, in my opinion, omit nothing that relates to the first part of our rule.

Now we must approach the second part of our task. That was to distinguish accurately the notions of simple things from those which are built up out of them; to see in both cases where falsity might come in, so that we might be on our guard and give our attention to those matters only in which certainty was possible. But here, as before, we must make certain assumptions which probably are not agreed on by all. It matters little, however, though they are not believed to be more real than those imaginary circles by means of which Astronomers describe their phenomena, provided that you employ them to aid you in discerning in each particular case what sort of knowledge is true and what false.

Finally, then, we assert that relatively to our knowledge single things should be taken in

an order different from that in which we should regard them when considered in their more real nature. Thus, for example, if we consider a body as having extension and figure, we shall indeed admit that from the point of view of the thing itself it is one and simple. For we cannot from that point of view regard it as compounded of corporeal nature, extension and figure, since these elements have never existed in isolation from each other. But relatively to our understanding we call it a compound constructed out of these three natures, because we have thought of them separately before we were able to judge that all three were found in one and the same subject. Hence here we shall treat of things only in relation to our understanding's awareness of them, and shall call those only simple, the cognition of which is so clear and so distinct that they cannot be analysed by the mind into others more distinctly known. Such are figure, extension, motion, etc.; all others we conceive to be in some way compounded out of these. This principle must be taken so universally as not even to leave out those objects which we sometimes obtain by abstraction from the simple natures themselves. This we do, for example, when we say that figure is the limit of an extended thing, conceiving by the term limit something more universal than by the term figure, since we can talk of a limit of duration, a limit of motion, and so on. But our contention is right, for then, even though we find the meaning of limit by abstracting it from figure, nevertheless it should not for that reason seem simpler than figure. Rather, since it is predicated of other things, as for example of the extreme bounds of a space of time or of a motion, etc., things which are wholly different from figure, it must be abstracted from those natures also; consequently it is something compounded out of a number of natures wholly diverse, of which it can be only ambiguously predicated.

Our second assertion is that those things which relatively to our understanding are called simple, are either purely intellectual or purely material, or else common both to intellect and to matter. Those are purely intellectual which our understanding apprehends by means of a certain inborn light, and without the aid of any corporeal image. That a number of such things exist is certain; and it is impossible to construct any corporeal idea which shall represent to us what the act of knowing is, what doubt is, what ignorance, and likewise what the action of the will is which it is possible to term volition, and so with other things. Yet we have a genuine knowledge of all these things, and know them so easily that in order to recognize them it is enough to be endowed with reason. Those things are purely material which we discern only in bodies; e.g. figure, extension, motion, etc. Finally those must be styled common which are ascribed now to corporeal things, now to spirits, without distinction. Such are existence, unity, duration and the like. To this group also we must ascribe those common notions which are, as it were, bonds for connecting together the other simple natures, and on whose evidence all the inferences which we obtain by reasoning depend. The following are examples:—things that are the same as a third thing are the same as one another. So too:—things which do not bear the same relation to a third thing, have some diversity from each other, etc. As a matter of fact these common notions can be discerned by the understanding either unaided or when it is aware of the images of material things.

But among these simple natures we must rank the privative and negative terms corresponding to them in so far as our intelligence grasps them. For it is quite as genuinely an act of knowledge by which I am intuitively aware of what nothing is, or an instant, or rest, as that by which I know what existence is, or lapse of time, or motion. This way of viewing the matter will be helpful in enabling us henceforth to say that all the rest of what we know is formed by composition out of these simple natures. Thus, for example, if I pronounce the judgment that some figure is not moving, I shall say that in a certain sense my idea is a complex of figure and rest; and so in other cases.

Thirdly, we assert that all these simple natures are known *per se* and are wholly free from falsity. It will be easy to show this, provided we distinguish that faculty of our understanding by which it has intuitive awareness of things and knows them, from that by which it judges, making use of affirmation and denial. For we may imagine ourselves to be ignorant of things which we really know, for example on such occasions as when we believe that in such things, over and above what we have present to us or attain to by thinking, there is something else hidden from us, and when this belief of ours is false. Whence it is evident that we are in error if we judge that any one of these simple natures is not completely known by us. For if our mind attains to the least acquain-

tance with it, as must be the case, since we are assumed to pass some judgment on it, this fact alone makes us infer that we know it completely. For otherwise it could not be said to be simple, but must be complex—a compound of that which is present in our perception of it, and that of which we think we are ignorant.

In the fourth place, we point out that the union of these things one with another is either necessary or contingent. It is necessary when one is so implied in the concept of another in a confused sort of way that we cannot conceive either distinctly, if our thought assigns to them separateness from each other. Thus figure is conjoined with extension, motion with duration or time, and so on, because it is impossible to conceive of a figure that has no extension, nor of a motion that has no duration. Thus likewise if I say "four and three are seven," this union is necessary. For we do not conceive the number seven distinctly unless we include in it the numbers three and four in some confused way. In the same way whatever is demonstrated of figures or numbers is necessarily united with that of which it is affirmed. Further, this necessity is not restricted to the field of sensible matters alone. The conclusion is necessary also in such a case—If Socrates says he doubts everything, it follows necessarily that he knows this at least—that he doubts. Likewise he knows that something can be either true or false, and so on, for all those consequences necessarily attach to the nature of doubt. The union, however, is contingent in those cases where the things are conjoined by no inseparable bond. Thus when we say a body is animate, a man is clothed, etc. Likewise many things are often necessarily united with one another, though most people, not noticing what their true relation is, reckon them among those that are contingently connected. As example, I give the following propositions:—"I exist, therefore God exists": also "I know, therefore I have a mind distinct from my body," etc. Finally, we must note that very many necessary propositions become contingent when converted. Thus, though from the fact that I exist I may infallibly conclude that God exists, it is not for that reason allowable to affirm that because God exists I also exist.

Fifthly, we remark that no knowledge is at any time possible of anything beyond those simple natures and what may be called their intermixture or combination with each other. Indeed it is often easier to be aware of several of them in union with each other, than to separate one of them from the others. For, to illustrate, I am able to know what a triangle is, though I have never thought that in that knowledge was contained the knowledge of an angle, a line, the number three, figure, extension, etc. But that does not prevent me from saying that the nature of the triangle is composed of all these natures, and that they are better known than the triangle since they are the elements which we comprehend in it. It is possible also that in the triangle many other features are involved which escape our notice, such as the magnitude of the angles, which are equal to two right angles, and the innumerable relations which exist between the sides and the angles, or the size of the area, etc.

Sixthly, we say that those natures which we call composite are known by us, either because experience shows us what they are, or because we ourselves are responsible for their composition. Matter of experience consists of what we perceive by sense, what we hear from the lips of others, and generally whatever reaches our understanding either from external sources or from that contemplation which our mind directs backwards on itself. Here it must be noted that no direct experience can ever deceive the understanding if it restrict its attention accurately to the object presented to it, just as it is given to it either at firsthand or by means of an image; and if it moreover refrain from judging that the imagination faithfully reports the objects of the senses, or that the senses take on the true forms of things, or in fine that external things always are as they appear to be; for in all these judgments we are exposed to error. This happens, for example, when we believe as fact what is merely a story that someone has told us; or when one who is ill with jaundice judges everything to be yellow because his eye is tinged with yellow. So finally, too, when the imagination is diseased, as in cases of melancholia, and a man thinks that his own disorderly fancies represent real things. But the understanding of a wise man will not be deceived by these fancies, since he will judge that whatever comes to him from his imagination is really depicted in it, but yet will never assert that the object has passed complete and without any alteration from the external world to his senses, and from his senses to his imagination, unless he has some previous ground for believing this. Moreover we ourselves are responsible for the composition of the things present to our understanding when we believe that there is something in them which our mind never experi-

ences when exercising direct perception. Thus if a man suffering from jaundice persuades himself that the things he sees are yellow, this thought of his will be composite, consisting partly of what his imagination represents to him, and partly of what he assumes on his own account, namely that the colour looks yellow not owing to the defect in his eye, but because the things he sees really are yellow. Whence the conclusion comes that we can go wrong only when the things we believe are in some way compounded by ourselves.

Seventhly, this compounding can come about in other ways, namely by impulse, by conjecture, or by deduction. Impulse sways the formation of judgments about things on the part of those whom their own initiative constrains to believe something, though they can assign no reason for their belief, but are merely determined either by some higher Power, or by their own free will, or by their fanciful disposition. The first cause is never a source of error, the second rarely, the third almost always. But a consideration of the first does not concern us here because it does not fall within the province of human skill. The working of conjecture is shown, for example, in this: water which is at a greater distance from the centre of the globe than earth, is likewise less dense substance, and likewise the air which is above the water, is still rarer; hence we hazard the guess that above the air nothing exists but a very pure aether, which is much rarer than air itself. Moreover nothing that we construct in this way really deceives us, if we merely judge it to be probable and never affirm it to be true; in fact it makes us better instructed.

Deduction is thus left to us as the only means of putting things together so as to be sure of their truth. Yet in it, too, there may be many defects. Thus if, in this space which is full of air, there is nothing to be perceived either by sight, touch, or any other sense, we conclude that the space is empty, we are in error, and our synthesis of the nature of a vacuum with that of this space is wrong. This is the result as often as we judge that we can deduce anything universal and necessary from a particular or contingent fact. But it is within our power to avoid this error, if, for example, we never interconnect any objects unless we are directly aware that the conjunction of the one with the other is wholly necessary. Thus we are justified if we deduce that nothing can have figure which has not extension, from the fact that figure and extension are necessarily conjoined.

From all these considerations we conclude firstly—that we have shown distinctly and, as we judge, by an adequate enumeration, what we were originally able to express only confusedly and in a rough and ready way. This was that mankind has no road towards certain knowledge open to it, save those of self-evident intuition and necessary deduction; further, that we have shown what those simple natures are of which we spoke in the eighth proposition. It is also quite clear that this mental vision extends both to all those simple natures, and to the knowledge of the necessary connections between them, and finally to everything else which the understanding accurately experiences either at first hand or in the imagination. Deduction, however, will be further treated in what follows.

Our second conclusion is that in order to know these simple natures no pains need be taken, because they are of themselves sufficiently well known. Application comes in only in isolating them from each other and scrutinizing them separately with steadfast mental gaze. There is no one whose intelligence is so dull as not to perceive that when he is seated he in some way differs from what he is when standing. But not everyone separates with equal distinctness the nature of position from the other elements contained in the cognition in question, or is able to assert that in this case nothing alters save the position. Now it is not without reason that we call attention to the above doctrine; for the learned have a way of being so clever as to contrive to render themselves blind to things that are in their own nature evident, and known by the simplest peasant. This happens when they try to explain by something more evident those things that are self-evident. For what they do is either to explain something else, or nothing at all. Who, for instance, does not perfectly see what that is, whatsoever it may be, in respect of which alteration occurs when we change position? But is there anyone who would grasp that very thing when he was told that *place is the surface of the body surrounding us*?[1] This would be strange seeing that that surface can change though I stay still and do not change my place, or that, on the contrary, it can so move along with me that, although it continues to surround me, I am nevertheless no longer in the same place. Do not these people really seem to use magic words which have a hidden force that eludes the grasp of human apprehension?

[1]Cf. reply to Obj. VI. (7), p. 228.

They define *motion*, a fact with which everyone is quite familiar, as *the actualisation of what exists in potentiality, in so far as it is potential!* Now who understands these words? And who at the same time does not know what motion is? Will not everyone admit that those philosophers have been trying to find a knot in a bulrush? We must therefore maintain that no definitions are to be used in explaining things of this kind lest we should take what is complex in place of what is simple. We must be content to isolate them from each other, and to give them, each of us, our individual attention, studying them with that degree of mental illumination which each of us possesses.

Our third conclusion is that the whole of human knowledge consists in a distinct perception of the way in which those simple natures combine in order to build up other objects. It is important to note this; because whenever some difficulty is brought forward for examination, almost everyone is brought to a standstill at the very outset, being in doubt as to the nature of the notions he ought to call to mind, and believing that he has to search for some new kind of fact previously unknown to him. Thus, if the question is, "what is the nature of the magnet?" people like that at once prognosticate difficulty and toil in the inquiry, and dismissing from mind every well-known fact, fasten on whatsoever is most difficult, vaguely hoping that by ranging over the fruitless field where multifarious causes lie, they will find something fresh. But he who reflects that there can be nothing to know in the magnet which does not consist of certain simple natures evident in themselves, will have no doubt how to proceed. He will first collect all the observations with which experience can supply him about this stone, and from these he will next try to deduce the character of that intermixture of simple natures which is necessary to produce all those effects which he has seen to take place in connection with the magnet. This achieved, he can boldly assert that he has discovered the real nature of the magnet in so far as human intelligence and the given experimental observations can supply him with this knowledge.

Finally, it follows fourthly from what has been said that we must not fancy that one kind of knowledge is more obscure than another, since all knowledge is of the same nature throughout, and consists solely in combining what is self-evident. This is a fact recognized by very few. People have their minds already occupied by the contrary opinion, and the more bold among them, indeed, allow themselves to uphold their private conjectures as though they were sound demonstrations, and in matters of which they are wholly ignorant feel premonitions of the vision of truths which seem to present themselves through a cloud. These they have no hesitation in propounding, attaching to their concepts certain words by means of which they are wont to carry on long and reasoned out discussions, but which in reality neither they nor their audience understand. On the other hand more diffident people often refrain from many investigations that are quite easy and are in the first degree necessary to life, merely because they think themselves unequal to the task. They believe that these matters can be discovered by others who are endowed with better mental faculties, and embrace the opinion of those in whose authority they have most confidence.

We assert fifthly that by deduction we can get only things from words, cause from effect, or effect from cause, like from like, or parts or the whole itself from the parts....

For the rest, in order that there may be no want of coherence in our series of precepts, we divide the whole matter of knowledge into simple propositions and "questions."[1] In connection with simple propositions the only precepts we give are those which prepare our cognitive faculties for fixing distinctly before them any objects, whatsoever they are, and scrutinizing them with keen intelligence, since propositions of this type do not arise as the result of inquiry, but present themselves to us spontaneously. This part of our task we have undertaken in the first twelve rules, in which, we believe, we have displayed everything which, in our opinion, can facilitate the exercise of our reason. But as to "questions" some of them can be perfectly well comprehended, even though we are ignorant of their solution; these we shall treat by themselves in the next twelve rules. Finally there are others, whose meaning is not quite clear, and these we reserve for the last twelve. This division has been made advisedly, both in order to avoid mentioning anything which presupposes an acquaintance with what follows, and also for the purpose of unfolding first what we feel to be most important first to inculcate in cultivating the mental powers. Among the "questions" whose

[1] *Quaestiones.* Quotation marks have been employed wherever it is important to remember Descartes' special technical use of this term.

meaning is quite plain, we must to begin with note that we place those only in which we perceive three things distinctly; to wit, the marks by which we can identify what we are looking for when it occurs; what precisely the fact is from which our answer ought to be deduced; and how it is to be proved that these (the ground and its consequence) so depend one on another that it is impossible for either to change while the other remains unchanged. In this way we shall have all the premisses we require, and the only thing remaining to be shown will be how to discover the conclusion. This will not be a matter of deducing some one fact from a single simple matter (we have already said that we can do this without the help of rules), but of disentangling so skilfully some one fact that is conditioned by a number of others which all involve one another, that in recognizing it there shall be no need to call upon a higher degree of mental power than in making the simplest inference. "Questions" of this kind, being highly abstract and occurring almost exclusively in Arithmetic and Geometry, seem to the inexperienced of little value. But I warn them that people ought to busy and exercise themselves a long time in learning this art, who desire to master the subsequent portions of this method, in which all the other types of "question" are treated

RULE XIII

Once a "question" is perfectly understood, we must free it of every conception superfluous to its meaning, state it in its simplest terms, and, having recourse to an enumeration, split it up into the various sections beyond which analysis cannot go in minuteness.

This is the only respect in which we imitate the Dialecticians; just as they, in teaching their doctrine of the forms of syllogism, assume that the terms or matter of their syllogisms are already known, so also we on this occasion lay it down as a prerequisite that the question to be solved should be perfectly understood. But we do not, as they, distinguish two extremes and a middle term. The following is the way in which we look at the whole matter. Firstly, in every "question" there must be something of which we are ignorant; otherwise there is no use asking the question. Secondly, this very matter must be disignated in some way or other; otherwise there would be nothing to determine us to investigate it rather than anything else. Thirdly, it can only be so designated by the aid of something else which is already known. All three conditions are realised even in questions that are not fully understood. Thus if the problem be the nature of the magnet, we already know what is meant by the two words "magnet" and "nature," and this knowledge determines us to seek one sort of answer rather than another, and so on. But over and above this, if the question is to be perfectly stated, we require that it should be wholly determinate, so that we shall have nothing more to seek for than what can be inferred from the data. For example, some one might set me the question, what is to be inferred about the nature of the magnet from that set of experiments precisely which Gilbert[1] asserts he has performed, be they trustworthy or not. So again the question may be, what my conclusion is as to the nature of sound, founding my judgment merely on the precise fact that the three strings A, B, and C give out an identical sound, when by hypothesis B, though twice as thick as A, but not longer, is kept in tension by a weight that is twice as heavy; while C, though no thicker than A, but merely twice as long, is nevertheless kept in tension by a weight four times as heavy. Other illustrations might be given; but they all make it quite clear how all imperfectly expressed "questions" may be reduced to others whose meaning is quite clear, as I shall show at greater length in the proper place. We see how it is possible to follow this rule in divesting any difficulty, where the problem is properly realised, of every superfluous conception, and in reducing it to a form in which we no longer deem that we are treating of this or that special matter, but are dealing only in a general way with certain magnitudes which have to be fitted together. Thus, to illustrate, after we have limited ourselves to the consideration of this or that set of experiments merely relative to the magnet, there is no difficulty in dismissing from view all other aspects of the case.

We add also that the problem ought to be reduced to its simplest statement in accordance with Rules V and VI, and resolved into parts in accordance with Rule VII. Thus if I employ a number of experiments in investigating the magnet, I shall run them over successively, taking each by itself. Again if my inquiry is about sound, as in the case above, I shall separately consider the relation between strings A and B, then that between A and C, and so on, so that afterwards my enumeration

[1]Presumably the English physicist W. Gilbert (1540–1603), author of *On the Loadstone.*

of results may be sufficient, and may embrace every case. These three rules are the only ones which the pure understanding need observe in dealing with the terms of any proposition before approaching its ultimate solution, though that requires us to employ the following eleven rules. The third part of this Treatise will show us more clearly how to apply them. Further by a "question" we understand everything in which either truth or falsity is found; and we must enumerate the different types of "question" in order to determine what we are able to accomplish in each case.

We have already said that there can be no falsity in the mere intuition of things, whether they are simple or united together. So conceived these are not called "questions," but they acquire that designation so soon as we prepare to pass some determinate judgment about them. Neither do we limit the title to those questions which are set us by other people. His own ignorance, or more correctly his own doubt, presented a subject of inquiry to Socrates when first he began to study it and to inquire whether it was true that he doubted everything, and maintained that such was indeed the case.

Moreover in our "questions" we seek to derive either things from words, or causes from effects, or effects from causes, or the whole or other parts from parts, or to infer several of these simultaneously.

We are said to seek to derive things from words when the difficulty consists merely in the obscurity of the language employed. To this class we refer firstly all riddles, like that of the Sphinx about the animal which to begin with is four-footed, then two-footed, and finally three-footed. A similar instance is that of the fishers who, standing on the bank with rods and hooks ready for the capture of fish, said that they no longer possessed those creatures which they had caught, but on the other hand those which they had not yet been able to catch. So in other cases; but besides these, in the majority of matters on which the learned dispute, the question is almost always one of names. We ought not to judge so ill of our great thinkers as to imagine that they conceive the objects themselves wrongly, in cases where they do not employ fit words in explaining them. Thus when people call *place* the *surface of the surrounding body*, there is no real error in their conception; they merely employ wrongly the word *place*, which by common use signifies that simple and self-evident nature in virtue of which a thing is said to be here or there. This consists wholly in a certain relation of the thing said to be in the place towards the parts of the space external to it, and is a feature which certain writers, seeing that the name place was reserved for the surface of the surrounding body, have improperly called the thing's *intrinsic position*. So it is in other cases; indeed these verbal questions are of such frequent occurrence, that almost all controversy would be removed from among Philosophers, if they were always to agree as to the meaning of words.

We seek to derive causes from effects when we ask concerning anything, whether it exists or what it is .

Since, however, when a "question" is propounded for solution we are frequently unable at once to discern its type, or to determine whether the problem is to derive things from words, or causes from effects, etc., for this reason it seems to be superfluous to say more here in detail about these matters. It will occupy less space and will be more convenient, if at the same time we go over in order all the steps which must be followed if we are to solve a problem of any sort. After that, when any "question" is set, we must strive to understand distinctly what the inquiry is about.

For frequently people are in such a hurry in their investigations, that they bring only a blank understanding to their solution, without having settled what the marks are by which they are to recognize the fact of which they are in search, if it chance to occur. This is a proceeding as foolish as that of a boy, who, sent on an errand by his master, should be so eager to obey as to run off without having received his orders or knowing where to go.

However, though in every "question" something must be unknown, otherwise there is no need to raise it, we should nevertheless so define this unknown element by means of specific conditions that we shall be determined towards the investigation of one thing rather than another. These conditions to which, we maintain, attention must be paid at the very outset. We shall succeed in this if we so direct our mental vision as to have a distinct and intuitive presentation of each by itself, and inquire diligently how far the unknown fact for which we are in search is limited by each. For the human mind is wont to fall into error in two ways here; it either assumes more than is really given in determining the question, or, on the other hand, leaves something out.

We must take care to assume neither more nor less than our data furnish us. This applies chiefly to riddles and other problems where the object of the skill employed is to try to puzzle people's wits. But frequently also we must bear it in mind in other "questions," when it appears as though we could assume as true for the purpose of their solution a certain matter which we have accepted, not because we had a good reason for doing so, but merely because we had always believed it. Thus, for example, in the riddle put by the Sphinx, it is not necessary to believe that the word "foot" refers merely to the real foot of an animal; we must inquire also whether the term cannot be transferred to other things, as it may be, as it happens, to the hands of an infant, or an old man's staff, because in either case these accessories are employed as feet are in walking. So too, in the fishermen's conundrum, we must beware of letting the thought of fish occupy our minds to the exclusion of those creatures which the poor so often carry about with them unwillingly, and fling away from them when caught. So again,we must be on our guard when inquiring into the construction of a vessel, such as we once saw, in the midst of which stood a column and upon that a figure of Tantalus in the attitude of a man who wants to drink. Water when poured into the vessel remained within without leaking as long as it was not high enough to enter the mouth of Tantalus; but as soon as it touched the unhappy man's lips the whole of it at once flowed out and escaped. Now at the first blush it seems as if the whole of the ingenuity consisted in the construction of this figure of Tantalus, whereas in reality this is a mere accompaniment of the fact requiring explanation, and in no way conditions it. For the whole difficulty consists solely in the problem of how the vessel was constructed so as to let out the whole of the water when that arrived at a certain height, whereas before none escaped. Finally, likewise, if we seek to extract from the recorded observations of the stars an answer to the question as to what we can assert about their motions, it is not to be gratuitously assumed that the earth is immoveable and established in the midst of the universe, as the Ancients would have it, because from our earliest years it appears to be so. We ought to regard this as dubious, in order afterwards to examine what certainty there is in this matter to which we are able to attain. So in other cases.

On the other hand we sin by omission when there is some condition requisite to the determination of the question either expressed in it or in some way to be understood, which we do not bear in mind. This may happen in an inquiry into the subject of perpetual motion, not as we meet with it in nature in the movements of the stars and the flowing of springs, but as a motion contrived by human industry. Numbers of people have believed this to be possible, their idea being that the earth is in perpetual motion in a circle round its own axis, while again the magnet retains all the properties of the earth. A man might then believe that he would discover a perpetual motion if he so contrived it that a magnet should revolve in a circle, or at least that it communicated its own motion along with its other properties to a piece of iron. Now although he were to succeed in this, it would not be a perpetual motion artificially contrived; all he did would be to utilize a natural motion, just as if he were to station a wheel in the current of a river so as to secure an unceasing motion on its part. Thus in his procedure he would have omitted a condition requisite for the resolution of his problem.

When we have once adequately grasped the meaning of a "question," we ought to try and see exactly wherein the difficulty consists, in order that, by separating it out from all complicating circumstances, we may solve it the more easily. But over and above this we must attend to the various separate problems involved in it, in order that if there are any which are easy to resolve we may omit them; when these are removed, only that will remain of which we are still in ignorance. Thus in that instance of the vessel which was described a short time ago, it is indeed quite easy to see how the vessel should be made; a column must be fixed in its centre, a bird[1] must be painted on it. But all these things will be set aside as not touching the essential point; thus we are left with the difficulty by itself, consisting in the fact that the whole of the water, which had previously remained in the vessel, after reaching a certain height, flows out. It is for this that we have to seek a reason.

Here therefore we maintain that what is worth while doing is simply this—to explore in an orderly way all the data furnished by the proposition, to set aside everything which we see is clearly immaterial, to retain what is necessarily bound up with the problem, and to reserve what is doubtful for a more careful examination.

[1]Translate "a valve must be fitted in it."

RULE XIV

The same rule is to be applied also to the real extension of bodies. It must be set before the imagination by means of mere figures, for this is the best way to make it clear to the understanding.

But in proposing to make use of the imagination as an aid to our thinking, we must note that whenever one unknown fact is deduced from another that is already known, that does not show that we discover any new kind of entity, but merely that this whole mass of knowledge is extended in such a way that we perceive that the matter sought for participates in one way or another in the nature of the data given in the proposition. For example if a man has been blind from his birth it is not to be expected that we shall be able by any train of reasoning to make him perceive the true ideas of the colours which we have derived from our senses. But if a man has indeed once perceived the primary colours, though he has never seen the intermediate or mixed tints, it is possible for him to construct the images of those which he has not seen from their likeness to the others, by a sort of deduction. Similarly if in the magnet there be any sort of nature the like of which our mind has never yet known, it is hopeless to expect that reasoning will ever make us grasp it; we should have to be furnished either with some new sense or with a divine intellect. But we shall believe ourselves to have attained whatever in this matter can be achieved by our human faculties, if we discern with all possible distinctness that mixture of entities or natures already known which produces just those effects which we notice in the magnet.

Indeed all these previously known entities, viz. extension, figure, motion and the like, the enumeration of which does not belong to this place, are recognized by means of an idea which is one and the same in the various subject matters. The figure of a silver crown which we imagine, is just the same as that of one that is golden. Further this common idea is transferred from one subject to another, merely by means of the simple comparison by which we affirm that the object sought for is in this or that respect like, or identical with, or equal to a particular datum. Consequently in every train of reasoning it is by comparison merely that we attain to a precise knowledge of the truth. Here is an example:—all A is B, all B is C, therefore all A is C. Here we compare with one another a *quaesitum* and a *datum*, viz. A and C, in respect of the fact that each is B, and so on. But because, as we have often announced, the syllogistic forms are of no aid in perceiving the truth about objects, it will be for the reader's profit to reject them altogether and to conceive that all knowledge whatsoever, other than that which consists in the simple and naked intuition of single independent objects, is a matter of the comparison of two things or more, with each other. In fact practically the whole of the task set the human reason consists in preparing for this operation; for when it is open and simple, we need no aid from art, but are bound to rely upon the light of nature alone, in beholding the truth which comparison gives us.

We must further mark that comparison should be simple and open, only as often as *quaesitum* and *datum* participate equally in a certain nature. Note that the only reason why preparation is required for comparison that is not of this nature is the fact that the common nature we spoke of does not exist equally in both, but is complicated with certain other relations or ratios. The chief part of our human industry consists merely in so transmuting these ratios as to show clearly a uniformity between the matter sought for and something else already known.

Next we must mark that nothing can be reduced to this uniformity, save that which admits of a greater and a less, and that all such matter is included under the term magnitude. Consequently when, in conformity with the previous rule, we have freed the terms of the problem from any reference to a particular subject, we shall discover that all we have left to deal with consists of magnitudes in general.

We shall, however, even in this case make use of our imagination, employing not the naked understanding but the intellect as aided by images of particulars depicted on the fancy. Finally we must note that nothing can be asserted of magnitudes in general that cannot also be ascribed to any particular instance.

This lets us easily conclude that there will be no slight profit in transferring whatsoever we find asserted of magnitudes in general to that particular species of magnitude which is most easily and distinctly depicted in our imagination. But it follows from what we stated about the twelfth rule that this must be the real extension of body abstracted from everything else except the fact that it has figure; for in that place we represented the imagination itself along with the ideas it contains as nothing

more than a really material body possessing extension and figure. This is also itself evident; for no other subject displays more distinctly differences in ratio of whatsoever kind. Though one thing can be said to be more or less white than another, or a sound sharper or flatter, and so on, it is yet impossible to determine exactly whether the greater exceeds the less in the proportion two to one, or three to one, etc., unless we treat the quantity as being in a certain way analogous to the extension of a body possessing figure. Let us then take it as fixed and certain that perfectly definite "questions" are almost free from difficulty other than that of transmuting ratios so that they may be stated as equations. Let us agree, too, that everything in which we discover precisely this difficulty, can be easily, and ought to be, disengaged from reference to every other subject, and immediately stated in terms of extension and figure. It is about these alone that we shall for this reason henceforth treat, up to and as far as the twenty-fifth rule, omitting the consideration of everything else.

My desire is that here I may find a reader who is an eager student of Arithmetic and Geometry, though indeed I should prefer him to have had no practice in these arts, rather than to be an adept after the ordinary standard. For the employment of the rules which I here unfold is much easier in the study of Arithmetic and Geometry (and it is all that is needed in learning them) than in inquiries of any other kind. Further, its usefulness as a means towards the attainment of a profounder knowledge is so great, that I have no hesitation in saying that it was not the case that this part of our method was invented for the purpose of dealing with mathematical problems, but rather that mathematics should be studied almost solely for the purpose of training us in this method. I shall presume no knowledge of anything in mathematics except perhaps such facts as are self-evident and obvious to everyone. But the way in which people ordinarily think about them, even though not vitiated by any glaring errors, yet obscures our knowledge with many ambiguous and ill-conceived principles, which we shall try incidentally to correct in the following pages.

By extension we understand whatever has length, breadth, and depth, not inquiring whether it be a real body or merely space; nor does it appear to require further explanation, since there is nothing more easily perceived by our imagination. Yet the learned frequently employ distinctions so subtle that the light of nature is dissipated in attending to them, and even those matters of which no peasant is ever in doubt become invested in obscurity. Hence we announce that by extension we do not here mean anything distinct and separate from the extended object itself; and we make it a rule not to recognize those metaphysical entities which really cannot be presented to the imagination. For even though someone could persuade himself, for example, that supposing every extended object in the universe were annihilated, that would not prevent extension in itself alone existing, this conception of his would not involve the use of any corporeal image, but would be based on a false judgment of the intellect working by itself. He will admit this himself, if he reflect attentively on this very image of extension when, as will then happen, he tries to construct it in his imagination. For he will notice that, as he perceives it, it is not divested of a reference to every object, but that his imagination of it is quite different from his judgment about it. Consequently, whatever our understanding may believe as to the truth of the matter, those abstract entities are never given to our imagination as separate from the objects in which they inhere.

But since henceforth we are to attempt nothing without the aid of the imagination, it will be worth our while to distinguish carefully the ideas which in each separate case are to convey to the understanding the meaning of the words we employ. To this end we submit for consideration these three forms of expression:—*extension occupies place*, *body possesses extension*, and *extension is not body*.

The first statement shows how extension may be substituted for that which is extended. My conception is entirely the same if I say *extension occupies place*, as when I say *that which is extended occupies place*. Yet that is no reason why, in order to avoid ambiguity, it should be better to use the term *that which is extended:* for that does not indicate so distinctly our precise meaning, which is, that a subject occupies place owing to the fact that it is extended. Someone might interpret the expression to mean merely *that which is extended is an object occupying place*, just in the same way as if I had said *that which is animate occupies place*. This explains why we announced that here we would treat of extension, preferring that to "the extended," although we believe that there is no difference in the conception of the two.

Let us now take up these words: *body possesses extension.* Here the meaning of *extension* is not identical with that of body, yet we do not construct two distinct ideas in our imagination, one of body, the other of extension, but merely a single image of extended body; and from the point of view of the thing it is exactly as if I had said: *body is extended,* or better, *the extended is extended.* This is a peculiarity of those entities which have their being merely in something else, and can never be conceived without the subject in which they exist. How different is it with those matters which are really distinct from the subjects of which they are predicated. If, for example, I say *Peter has wealth,* my idea of Peter is quite different from that of wealth. So if I say *Paul is wealthy,* my image is quite different from that which I should have if I said *the wealthy man is wealthy.* Failure to distinguish the diversity between these two cases is the cause of the error of those numerous people who believe that extension contains something distinct from that which is extended, in the same way as Paul's wealth is something different from Paul himself.

Finally, take the expression: *extension is not body.* Here the term extension is taken quite otherwise than as above. When we give it this meaning there is no special idea corresponding to it in the imagination. In fact this entire assertion is the work of the naked understanding, which alone has the power of separating out abstract entities of this type. But this is a stumbling-block for many, who, not perceiving that extension so taken, cannot be grasped by the imagination, represent it to themselves by means of a genuine image. Now such an idea necessarily involves the concept of body, and if they say that extension so conceived is not body, their heedlessness involves them in the contradiction of saying that *the same thing is at the same time body and not body.* It is likewise of great moment to distinguish the meaning of the enunciations in which such names as *extension, figure, number, superficies, line, point, unity,* etc. are used in so restricted a way as to exclude matters from which they are not really distinct. Thus when we say: *extension* or *figure is not body; number is not the thing that is counted; a superficies is the boundary of a body, the line the limit of a surface, the point of a line; unity is not a quantity,* etc.; all these and similar propositions must be taken altogether outside the bounds of the imagination, if they are to be true. Consequently we shall not discuss them in the sequel.

But we should carefully note that in all other propositions in which these terms, though retaining the same signification and employed in abstraction from their subject matter, do not exclude or deny anything from which they are not really distinct, it is both possible and necessary to use the imagination as an aid. The reason is that even though the understanding in the strict sense attends merely to what is signified by the name, the imagination nevertheless ought to fashion a correct image of the object, in order that the very understanding itself may be able to fix upon other features belonging to it that are not expressed by the name in question, whenever there is occasion to do so, and may never imprudently believe that they have been excluded. Thus, if number be the question, we imagine an object which we can measure by summing a plurality of units. Now though it is allowable for the understanding to confine its attention for the present solely to the multiplicity displayed by the object, we must be on our guard nevertheless not on that account afterwards to come to any conclusion which implies that the object which we have described numerically has been excluded from our concept. But this is what those people do who ascribe mysterious properties to number, empty inanities in which they certainly would not believe so strongly, unless they conceived that number was something distinct from the things we number. In the same way, if we are dealing with figure, let us remember that we are concerned with an extended subject, though we restrict ourselves to conceiving it merely as possessing figure. When body is the object let us reflect that we are dealing with the very same thing, taken as possessing length, breadth and depth. Where superficies comes in, our object will still be the same though we conceive it as having length and breadth, and we shall leave out the element of depth, without denying it. The line will be considered as having length merely, while in the case of the point the object, though still the same, will be divested in our thought of every characteristic save that of being something existent.

In spite of the way in which I have dwelt on this topic, I fear that men's minds are so dominated by prejudice that very few are free from the danger of losing their way here, and that, notwithstanding the length of my discourse, I shall be found to have explained myself too briefly. Those very disciplines Arithmetic and Geometry, though the most certain of all the

sciences, nevertheless lead us astray here. For is there a single Arithmetician who does not believe that the numbers with which he deals are not merely held in abstraction from any subject matter by the understanding, but are really distinct objects of the imagination? Does not your Geometrician obscure the clearness of his subject by employing irreconcileable principles? He tells you that lines have no breadth, surfaces no depth; yet he subsequently wishes to generate the one out of the other, not noticing that a line, the movement of which is conceived to create a surface, is really a body; or that, on the other hand, the line which has no breadth, is merely a mode of body. But, not to take more time in going over these matters, it will be more expeditious for us to expound the way in which we assume our object should be taken, in order that we may most easily give a proof of whatsoever is true in Arithmetic and Geometry.

Here therefore we deal with an extended object, considering nothing at all involved in it save extension, and purposely refraining from using the word quantity, because there are certain Philosophers so subtle as to distinguish it also from extension. We assume such a simplification of our problems as to leave nothing else to be inquired about except the determination of a certain extension by comparing it with a certain other extension that is already determinately known. For here we do not look to discover any new sort of fact; we merely wish to make a simplification of ratios, be they ever so involved, such that we may discover some equation between what is unknown and something known. Since this is so, it is certain that whatsoever differences in ratio exist in these subjects can be found to prevail also between two or more extensions. Hence our purpose is sufficiently served if in extension itself we consider everything that can aid us in setting out differences in ratio; but there are only three such features, viz. dimension, unity and figure.

By dimension I understand nothing but the mode and aspect according to which a subject is considered to be measurable. Thus it is not merely the case that length, breadth and depth are dimensions; but weight also is a dimension in terms of which the heaviness of objects is estimated. So, too, speed is a dimension of motion, and there are an infinite number of similar instances. For that very division of the whole into a number of parts of identical nature, whether it exist in the real order of things or be merely the work of the understanding, gives us exactly that dimension in terms of which we apply number to objects. Again that mode which constitutes number is properly said to be a species of dimension, though there is not an absolute identity between the meaning of the two terms. For if we proceed by taking part after part until we reach the whole, the operation is then said to be counting, whereas if conversely we look upon the whole as something split up into parts, it is an object which we measure. Thus we measure centuries by years, days, hours and moments, while if we count up moments, hours, days and years, we shall finish with a total of centuries.

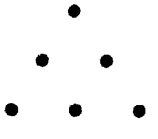
It clearly follows that there may be an infinite number of dimensions in the same subject, which make no addition at all to the objects which possess them, but have the same meaning whether they are based on anything real in the objects themselves, or are the arbitrary inventions of our own mind. Weight is indeed something real existing in a body, and the speed of motion is a reality, and so with the division of a century into years and days. But it is otherwise with the division of the day into hours and moments, etc. Yet all these subdivisions are exactly similar if considered merely from the point of view of dimension, as we ought to regard them both here and in the science of Mathematics. It falls rather to Physics to inquire whether they are founded on anything real.

Recognition of this fact throws much light on Geometry, since in that science almost everyone goes wrong in conceiving that quantity has three species, the line, the superficies, and the solid. But we have already stated that the line and the superficies are not conceived as being really distinct from solid body, or from one another. Moreover if they are taken in their bare essence as abstractions of the understanding, they are no more diverse species of quantity than the "animal" and "living creature" in man are diverse species of substance. Incidentally also we have to note that the three dimensions of body, length, breadth and depth, are only in name distinct from one another. For there is nothing to prevent us, in any solid body with which we are dealing, from taking any of the extensions it presents as the length, or any other as its depth, and so on. And though these three dimensions have a real basis in every extended object quâ extended, we have nevertheless no special concern in this science with them more than with countless

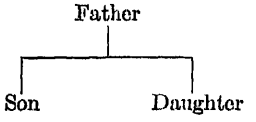
others, which are either mental creations or have some other ground in objects. For example in the case of the triangle, if we wish to measure it exactly, we must acquaint ourselves with three features of its existence, viz. either its three sides, or two sides and an angle, or two angles and its area, and so forth. Now these can all be styled dimensions. Similarly in a trapezium five facts have to be noted, in a tetrahedron six, and so on. But if we wish to choose here those dimensions which shall give most aid to our imagination, we shall never attend at the same time to more than one or two of those depicted in our imagination, even though we know that in the matter set before us with which we are dealing several others are involved. For the art of our method consists in distinguishing as many elements as possible, so that though we attend to only a few simultaneously, we shall yet cover them all in time, taking one after the other.

The unit is that common element in which, as above remarked, all the things compared with each other should equally participate. If this be not already settled in our problems, we can represent it by one of the magnitudes already presented to us, or by any other magnitude we like, and it will be the common measure of all the others. We shall understand that in it there exists every dimension found in those very widely sundered facts which are to be compared with each other, and we shall conceive it either (1) merely as something extended, omitting every other more precise determination—and then it will be identical with the point of Geometry, considered as generating a line by its movement; or (2) we shall conceive it as a line, or (3) as a square.

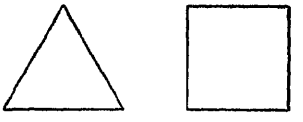
To come to figures, we have already shown above how it is they alone that give us a means of constructing the images of all objects whatsoever. It remains to give notice in this place, that of the innumerable diverse species of figure, we shall employ only those which most readily express differences of relation or proportion. Moreover there are two sorts of objects only which are compared with each other, viz. numerical assemblages and magnitudes. Now there are also two sorts of figures by means of which these may be presented to our conception. For example we have the points



which represent a triangular[1] number, or again the "tree" which illustrates genealogical relation as in such a case—



So in similar instances. Now these are figures designed to express numerical assemblages; but those which are continuous and undivided like the triangle, the square, etc.,



explain the nature of magnitudes.

But in order that we may point out which of all these figures we are going to use, it ought to be known that all the relations which can exist between things of this kind, must be referred to two heads, viz. either to order or to measurement.

We must further realise that while the discovery of an order is no light task, as may be seen throughout this treatise, which makes this practically its sole subject, yet once the order has been discovered there is no difficulty at all in knowing it. The seventh rule shows us how we may easily review in sequence mentally the separate elements which have been arranged in order, for the reason that in this class of relation the bond between the terms is a direct one involving nothing but the terms themselves, and not requiring mediation by means of a third term, as is the case in measurement. The unfolding of relations of measurement will therefore be all that we shall treat of here. For I recognize the order in which A and B stand, without considering anything except these two—the extreme terms of the relation. But I can recognize the ratio of the magnitude of two to that of three, only by considering some third thing, namely unity, which is the common measure of both.

We must likewise bear in mind that, by the help of the unit we have assumed, continuous

[1] "Triangular" numbers are the sums of the natural numbers, viz. 1, 3, 6, 10, etc., and thus can be constructed from any number n according to the formula $\frac{n(n+1)}{2}$.

magnitudes can sometimes be reduced in their entirety to numerical expressions, and that this can always be partly realised. Further it is possible to arrange our assemblage of units in such an order that the problem which previously was one requiring the solution of a question in measurement, is now a matter merely involving an inspection of order. Now our method helps us greatly in making the progress which this transformation effects.

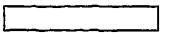
Finally, remember that of the dimensions of continuous magnitude none are more distinctly conceived than length and breadth, and that we ought not to attend to more than these two simultaneously in the same figure, if we are to compare two diverse things with each other. The reason is, that when we have more than two diverse things to compare with each other, our method consists in reviewing them successively and attending only to two of them at the same time.

Observation of these facts leads us easily to our conclusion. This is that there is no less reason for abstracting our propositions from those figures of which Geometry treats, if the inquiry is one involving them, than from any other subject matter. Further, in doing so we need retain nothing but rectilinear and rectangular superficies, or else straight lines, which we also call figures, because they serve quite as well as surfaces in aiding us to imagine an object which actually has extension, as we have already said. Finally those same figures have to represent for us now continuous magnitudes, again a plurality of units or number also. Human ingenuity can devise nothing simpler for the complete expression of differences of relation.

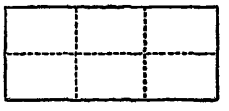
RULE XV

It is likewise very often helpful to draw these figures and display them to the external senses, in order thus to facilitate the continued fixation of our attention.

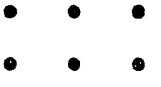
THE way in which these figures should be depicted so that, in being displayed before our eyes, the images may be the more distinctly formed in our imagination is quite self-evident. To begin with we represent unity in three ways, viz. by a square, □, if we consider our unit as having length and breadth, or secondly by a line, ——, if we take it merely as having length, or lastly by a point, •, if we think only of the fact that it is that by aid of which we construct a numerical assemblage. But however it is depicted and conceived, we shall always remember that the unit is an object extended in every direction, and admitting of countless dimensions. So also the terms of our proposition, in cases where we have to attend at the same time to two different magnitudes belonging to them, will be represented by a rectangle whose two sides will be the two magnitudes in question. Where they are incommensurable with our unit we shall employ the following figure,

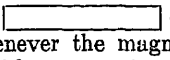


but where they are commensurable we shall use this



or this



and nothing more is needed save where it is a question of a numerical assemblage of units. Finally if we attend only to one of the magnitudes of the terms employed, we shall portray that either as a rectangle, of which one side is the magnitude considered and the other is unity, thus  — and this will happen whenever the magnitude has to be compared with some surface. Or we shall employ a line alone, in this fashion, ——, if we take it as an incommensurable length; or thus, • • • • •, if it be a number.

RULE XVI

When we come across matters which do not require our present attention, it is better, even though they are necessary to our conclusion, to represent them by highly abbreviated symbols, rather than by complete figures. This guards against error due to defect of memory on the one hand, and, on the other, prevents that distraction of thought which an effort to keep those matters in mind while attending to other inferences would cause.

BUT because our maxim is that not more than two different dimensions out of the countless number that can be depicted in our imagination ought to be the object either of our bodily or of our mental vision, it is of importance so to retain all those outside the range of present attention that they may easily come up to mind as often as need requires.

Now memory seems to be a faculty created by nature for this very purpose. But since it is liable to fail us, and in order to obviate the need of expending any part of our attention in refreshing it, while we are engaged with other thoughts, art has most opportunely invented the device of writing. Relying on the help this gives us, we leave nothing whatsoever to memory, but keep our imagination wholly free to receive the ideas which are immediately occupying us, and set down on paper whatever ought to be preserved. In doing so we employ the very briefest symbols, in order that, after distinctly examining each point in accordance with Rule IX, we may be able, as Rule XI bids us do, to traverse them all with an extremely rapid motion of our thought and include as many as possible in a single intuitive glance.

Everything, therefore, which is to be looked upon as single from the point of view of the solution of our problem, will be represented by a single symbol which can be constructed in any way we please. But to make things easier we shall employ the characters a, b, c, etc. for expressing magnitudes already known, and A, B, C, etc. for symbolising those that are unknown. To these we shall often prefix the numerical symbols, 1, 2, 3, 4, etc., for the purpose of making clear their number, and again we shall append those symbols to the former when we want to indicate the number of the relations which are to be remarked in them. Thus if I employ the formula $2a^3$ that will be the equivalent of the words "the double of the magnitude which is symbolised by the letter a, and which contains three relations." By this device not only shall we economize our words, but, which is the chief thing, display the terms of our problem in such a detached and unencumbered way that, even though it is so full as to omit nothing, there will nevertheless be nothing superfluous to be discovered in our symbols, or anything to exercise our mental powers to no purpose, by requiring the mind to grasp a number of things at the same time.

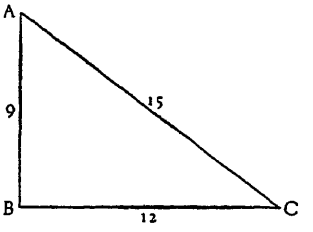
In order that all this may be more clearly understood, we must note first, that while Arithmeticians have been wont to designate undivided magnitudes by groups of units, or else by some number, we on the other hand abstract at this point from numbers themselves no less than from Geometrical figures or anything else, as we did a little time ago. Our reason for doing this is partly to avoid the tedium of a long and superfluous calculation, but chiefly that those portions of the matter considered which are relevant to the problem may always remain distinct, and may not be entangled with numbers that are of no help to us at all. Thus if we are trying to find the hypotenuse of the right-angled triangle whose sides are 9 and 12, the Arithmetician will tell us that it is $\sqrt{225}$, i.e. 15. But we shall write a and b in place of 9 and 12, and shall find the hypotenuse to be $\sqrt{a^2+b^2}$; and the two members of the expression a^2 and b^2 will remain distinct, whereas the number confuses them altogether.

Note further that by the number of relations attaching to a quantity I mean a sequence of ratios in continued proportion, such as the Algebra now in vogue attempts to express by sundry dimensions and figures. It calls the first of these the radix, the second the square, the third the cube, the fourth the biquadratic, and so on. I confess that for a long time I myself was imposed upon by these names. For, after the straight line and the square there was nothing which seemed to be capable of being placed more clearly before my imagination than the cube and the other figures of the same type; and with their aid I succeeded in solving not a few difficulties. But at last, after testing the matter well, I discovered that I had never found out anything by their means which I could not have recognized more easily and distinctly without employing their aid. I saw that this whole nomenclature must be abandoned, if our conceptions are not to become confused; for that very magnitude which goes by the name of the cube or the biquadratic, is nevertheless never to be presented to the imagination otherwise than as a line or a surface, in accordance with the previous rule. We must therefore be very clear about the fact that the radix, the square, the cube, etc., are merely magnitudes in continued proportion, which always imply the previous assumption of that arbitrarily chosen unit of which we spoke above. Now the first proportional is related to this unit directly and by a single ratio. But the second proportional requires the mediation of the first, and consequently is related to the unit by a pair of ratios. The third, being mediated by the first and second, has a triple relation to the standard unit, and so on. Therefore we shall henceforth call that magnitude, which in Algebra is styled the radix, the first proportional; that called the square we shall term the second proportional, and so in other cases.

Finally it must be noticed that even though here, in order to examine the nature of a difficulty, we abstract the terms involved from certain numerical complications, it yet often happens that a simpler solution will be found by employing the given numbers than if we abstract from them. This is due to the double function of numbers, already pointed out, which use the same symbols to express now order, and now measure. Hence, after seeking a solution in general terms for our problem, we ought to transform its terms by substituting for them the given numbers, in order to see whether these supply us with any simpler solution. Thus, to illustrate, after seeing that the hypotenuse of the right-angled triangle whose sides are a and b is $\sqrt{a^2+b^2}$, we should substitute 81 for a^2, and 144 for b^2. These added together give 225, the root of which, or mean proportional between unity and 225, is 15. This will let us see that a hypotenuse whose length is 15 is commensurable with sides whose lengths are 9 and 12, quite apart from the general law that it is the hypotenuse of a right-angled triangle whose sides are as 3 to 4. We, whose object is to discover a knowledge of things which shall be evident and distinct, insist on all those distinctions. It is quite otherwise with Arithmeticians, who, if the result required turns up, are quite content even though they do not perceive how it depends upon the data, though it is really in knowledge of this kind alone that science properly consists.

Moreover, it must be observed that, as a general rule, nothing that does not require to be continuously borne in mind ought to be committed to memory, if we can set it down on paper. This is to prevent that waste of our powers which occurs if some part of our attention is taken up with the presence of an object in our thought which it is superfluous to bear in mind. What we ought to do is to make a reference-table and set down in it the terms of the problem as they are first stated. Then we should state the way in which the abstract formulation is to be made and the symbols to be employed, in order that, when the solution has been obtained in terms of these symbols, we may easily apply it, without calling in the aid of memory at all, to the particular case we are considering: for it is only in passing from a lesser to a greater degree of generality that abstraction has any *raison d'être*. What I should write therefore would be something like this:—

In the right-angled triangle ABC to find the hypotenuse AC (stating the problem abstractly, in order that the derivation of the length of



the hypotenuse from the lengths of the sides may be quite general). Then for AB, which is equal to 9, I shall substitute a; for BC, equal to 12, I put b, and similarly in other cases.

To conclude, we draw attention to the fact that these four rules will be further employed in the third part of this Treatise, though we shall conceive them somewhat more generally than we have been doing. But all this will be explained in its proper place.

RULE XVII

When a problem is proposed for discussion we should run it over, taking a direct course, and for this reason neglecting the fact that some of its terms are known, others unknown. To follow the true connection, when presenting to mind the dependence of separate items on one another, will also aid us to do this.

THE four previous rules showed us how, when the problems are determinate and fully comprehended, we may abstract them from their subject matter and so transform them that nothing remains to be investigated save how to discover certain magnitudes, from the fact that they bear such and such a relation to certain other magnitudes already given. But in the five following rules we shall now explain how these same problems are to be treated in such a way that though a single proposition contains ever so many unknown magnitudes they may all be subordinated to one another; the second will stand to the first, as the first to unity, and so too the third to the second, and the fourth to the third, and so in succession, making, however numerous, a total magnitude equal to a certain known magnitude. In doing this our method will be so sure that we may safely affirm that it passes the wit of man to reduce our terms to anything simpler.

For the present, however, I remark that in every inquiry that is to be solved by deduction there is one way that is plain and direct, by which we may more easily than by any other

pass from one set of terms to another, while all other routes are more difficult and indirect. In order to understand this we must remember what was said relative to the eleventh rule, where we expounded the nature of that chain of propositions, a comparison of the neighbouring members of which enables us to see how the first is related to the last, even though it is not so easy to deduce the intermediate terms from the extremes. Now therefore if we fix our attention on the interdependence of the various links, without ever interrupting the order, so that we may thence infer how the last depends upon the first, we review the problem in a direct manner. But, on the other hand, if, from the fact that we know the first and the last to be connected with each other in a certain way, we should want to deduce the nature of the middle terms which connect them, we should then be following an order that was wholly indirect and upside down. But because here we are considering only involved inquiries, in which the problem is, given certain extremes, to find certain intermediates by the inverse process of reasoning, the whole of the device here disclosed will consist in treating the unknown as though they were known, and thus being able to adopt the easy and direct method of investigation even in problems involving any amount of intricacy. There is nothing to prevent us always achieving this result, since we have assumed from the commencement of this section of our work that we recognize the dependence of the unknown terms in the inquiry on those that are known to be such that the former are determined by the latter. This determination also is such that if, recognizing it, we consider the terms which first present themselves and reckon them even though unknown among the known, and thus deduce from them step by step and by a true connection all the other terms, even those which are known, treating them as though they were unknown, we shall fully realise the purpose of this rule. Illustrations of this doctrine, as of the most of what is immediately to follow, will be reserved until the twenty-fourth rule[1], since it will be more convenient to expound them there.

RULE XVIII

To this end only four operations are required, addition, subtraction, multiplication and division. Of these the two latter are often to be dispensed with here, both in order to avoid any unforeseen complication, and because it will be easier to deal with them at a later stage.

It is often from lack of experience on the part of the teacher that the multiplicity of rules proceeds; and matters that might have been reduced to one general rule are less clear if distributed among many particular statements. Wherefore we propose to reduce the whole of the operations which it is advisable to employ in going through our inquiry, i.e. in deducing certain magnitudes from others, to as few as four heads. It will become clear when we come to explain these how it is that they suffice for the purpose.

This is how we proceed. If we arrive at the knowledge of one magnitude owing to the fact that we already know the parts of which it is composed, the process is one of addition. If we discover the part because we already know the whole and the excess of the whole over this part, it is division. Further, it is impossible to derive a magnitude from others that are determinately fixed, and in which it is in any way contained, by any other methods. But if we have to derive a magnitude from others from which it is wholly diverse and in which it is in nowise contained, we must find some other way of relating it to them. Now if we trace out this connection or relation directly we must employ multiplication; if indirectly, division.

In explaining clearly these latter two operations the fact must be grasped that the unit of which we spoke before is here the basis and foundation of all the relations, and has the first place in the series of magnitudes in continued proportion. Further, remember that the given magnitudes occupy the second position, while those to be discovered stand at the third, the fourth and the remaining points in the series, if the proportion be direct. If, however, the proportion be indirect, the magnitude to be discovered occupies the second position or the other intermediate points, and that which is given, the last.

Thus if it is stated that as unity is to a, say to 5, which is given, so is b, i.e. 7, to the magnitude to be found[2], which is ab, i.e. 35, then a and b are at the second position, and ab, their product, at the third. So too if we are further told that as 1 is to c, say 9, so is ab, say 35, to the magnitude we are seeking, i.e. 315, then abc is in the fourth position, and is the product

[1] No such rule has been found among Descartes' papers.

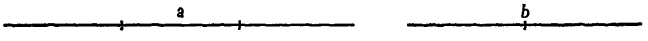
[2] Note that here Descartes does not, and could not conveniently, adhere to his scheme of employ

of two multiplications among the terms *a*, *b* and *c*, which are at the second position; so it is in other cases. Likewise as 1 is to *a*, say 5, so *a*, i.e. 5, is to a^2, i.e. 25. Again, as unity is to *a*, i.e. 5, so is a^2, i.e. 25, to a^3, i.e. 125; and finally as unity is to *a*, i.e. 5, so is a^3, i.e. 125, to a^4, i.e. 625, and so on. For the multiplication is performed in precisely the same way, whether the magnitude is multiplied by itself or by some other quite different number.

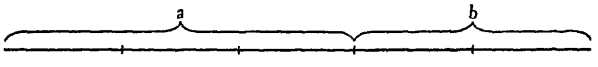
But if we now are told that, as unity is to *a*, say 5, the given divisor, so is *B*, say 7, *the quaesitum*, to *ab*, i.e. 35, the given dividend, we have on this occasion an example of the indirect or inverted order. For the only way to discover *B*, the *quaesitum*, is to divide the given *ab* by *a*, which is also given. The case is the same if the proposition is, "as unity is to *A*, say 5, the *quaesitum*, so is this *A* to a^2, i.e. 25, which is given"; or again, "as unity is to *A*, i.e. 5, the *quaesitum*, so is A^2, i.e. 25, which we also have to discover, to a^3, i.e. 125, which is given"; similarly in other cases. All these processes fall under the title "division," although we must note that these latter specimens of the process contain more difficulty than the former, because the magnitude to be found comes in a greater number of times in them, and consequently it involves a greater number of relations in such problems. For on such occasions the meaning is the same as if the enunciation were, "extract the square root of a^2, i.e. 25," or "extract the cube root of a^3, i.e. 125," and so in other cases. This then is the way in which Arithmeticians commonly put the matter. But alternatively we may explain the problems in the terms employed by Geometricians: it comes to the same thing if we say, "find a mean proportional between that assumed magnitude, which we call unity, and that indicated by a^2," or "find two mean proportionals between unity and a^3," and so in other cases.

From these considerations it is easy to infer how these two operations suffice for the discovery of any magnitudes whatsoever which are to be deduced from others in virtue of some relation. And now that we have grasped them, the next thing to do is to show how these operations are to be brought before the scrutiny of the imagination and how presented to our actual vision, in order that we may explain how they may be used or practised.

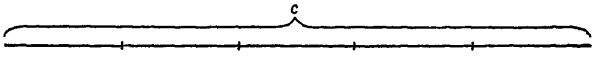
In addition or subtraction we conceive our object under the aspect of a line, or of some extended magnitude in which length is alone to be considered. For if we are to add line *a* to line *b*,



we add the one to the other in the following way *ab*,



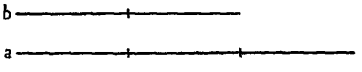
and get as a result *c*.



But if the smaller has to be taken from the larger, viz. *b* from *a*,



we place the one above the other thus,

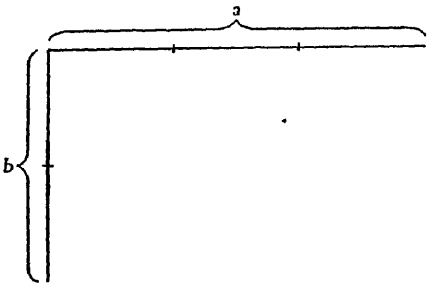


and this will give us that part of the larger which the smaller cannot cover, viz. ——.

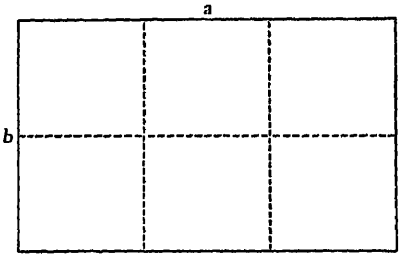
In multiplication we also conceive the given magnitudes as lines. But we imagine a rectangle to be constructed out of them; for, if we multiply *a* by *b*,



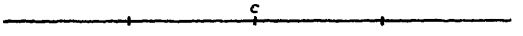
we fit them together at right angles in the following way,



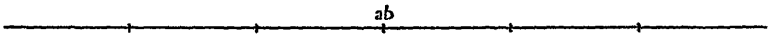
and so make the rectangle



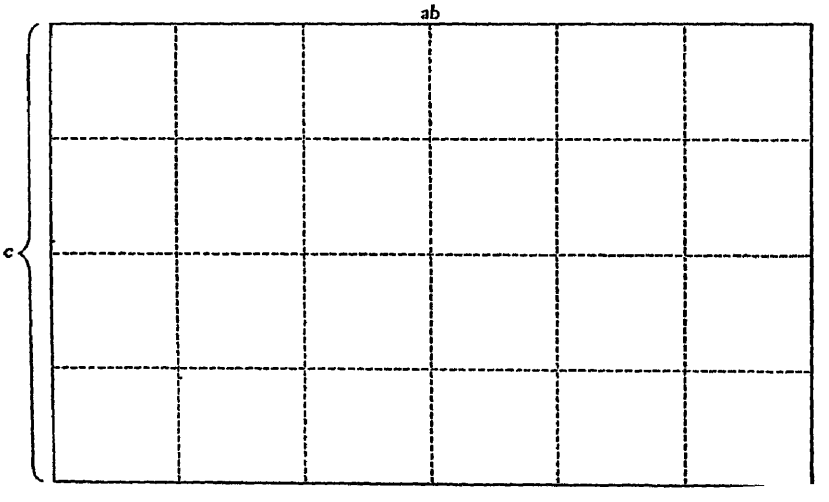
Again, if we wish to multiply *ab* by *c*,



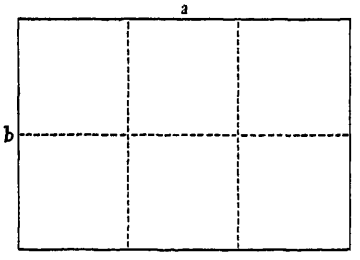
we ought to conceive *ab* as a line, viz. *ab*,



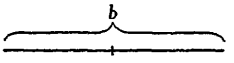
in order that to represent *abc* we may obtain the following figure:



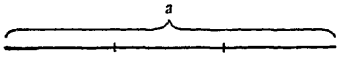
Finally, in a division in which the divisor is given, we imagine the magnitude to be divided to be a rectangle, one side of which is the divisor and the other the quotient. Thus if the rectangle *ab* is to be divided by *a*,



we take away from it the breadth *a* and are left with *b* for quotient:



On the contrary. if this rectangle is divided by *b*, we take away the height *b*, and the quotient will be *a*,



But in those divisions in which the divisor is not given, but only indicated by some relation, as when we are bidden extract the square or cube root, then we must note that the term to be divided and all the others must be always conceived as lines in continued proportion, of which the first is unity, the last the magnitude to be divided. The way in which any number of mean proportionals between this and unity may be discovered will be disclosed in its proper place. At present it is sufficient to have pointed out that according to our hypothesis those operations have not yet been fully dealt with here, since to be carried out they require an indirect and reverse movement on the part of the imagination; and at present we are treating only of questions in which the movement of thought is to be direct.

As for the other[1] operations, they can be carried out with the greatest ease in the way in which we have stated they are to be conceived. Nevertheless it remains for us to show how the terms employed in them are to be constructed. For even though on our first taking up some problem we are free to conceive the terms involved as lines or as rectangles, without introducing any other figures, as was said in reference to the fourteenth rule, nevertheless it is frequently the case that, in the course of the solution, what was a rectangle, constructed by the multiplication of two lines, must presently be conceived as a line, for the purpose of some further operation. Or it may be the case that the same rectangle, or a line formed by some addition or subtraction, has next to be conceived as some other rectangle drawn upon the line by which it is to be divided.

It is therefore worth our while here to expound how every rectangle may be transformed into a line, and conversely how a line or even a rectangle may be turned into another rectangle of which the side is indicated. This is the easiest thing in the world for Geometricians to do, provided they recognize that whenever we compare lines with some rectangle, as here, we always conceive those lines as rectangles, one side of which is the length that we took to represent our unit. For if we do so the whole matter resolves itself into the following proposition: Given a rectangle, to construct another rectangle equal to it upon a given side.

Now though this problem is one familiar to a mere beginner in Geometry, I wish to explain it, lest I should seem to have omitted something.

RULE XIX

Employing this method of reasoning we have to find out as many magnitudes as we have unknown terms, treated as though they were known,

[1]The direct operations.

for the purpose of handling the problem in the direct way; and these must be expressed in the two different ways. For this will give us as many equations as there are unknowns.

RULE XX

Having got our equations, we must proceed to carry out such operations as we have neglected, taking care never to multiply where we can divide.

RULE XXI

If there are several equations of this kind, we should reduce them all to a single one, viz. that the terms of which do not occupy so many places in the series of magnitudes that are in continued proportion. The terms of the equation should then be themselves arranged in the order which this series follows.